Part IB Physics

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Contents

7 Condensed Matter Physics

7.1 Before Everything in Condensed Matter Physics

The condensed matter physics (CMP) course in Part IB is the shortest introduction possible to solid state and quantum matters. It covers some chapters in Kittel's famous book [1], but not all due to time limitations. If one is looking for some fundamental knowledge in matter and materials, they should look for some other courses.

One should already learnt how to describe the structure of crystals: the 14 three-dimensional Bravais lattices, the conventional and primitive unit cells, Miller index, etc.

Because this is what we learned in the IA materials science course. Let's remind ourselves that

$$[hkl] \quad \langle hkl\rangle \quad (hkl) \quad \{hkl\} \tag{7.1}$$

denotes a single direction, a set of directions, a single plane, a set of planes, respectively.

7.2 Reciprocal Lattices and Diffraction

Generally, for any Bravais lattices with basis vectors $\vec{a}, \vec{b}, \vec{c}$, we can construct the relative position from any lattice point to another as

$$\vec{r} = u\vec{a} + v\vec{b} + w\vec{c}. \tag{7.2}$$

The reciprocal lattice, which was defined mathematically in part IA mathematics, is[1]

$$\vec{A} = \frac{2\pi\vec{b}\times\vec{c}}{[\vec{a},\vec{b},\vec{c}]}, \tag{7.3}$$

where the 2π factor is there for normalisation, so we have properties like

$$\vec{A}\cdot\vec{a} = 2\pi; \qquad \vec{A}\cdot\vec{b} = \vec{A}\cdot\vec{c} = 0, \quad \text{etc.} \tag{7.4}$$

Information is now totally transferred into the reciprocal lattice. The position from one lattice point to another is now

$$\vec{G}_{hkl} = h\vec{A} + k\vec{B} + l\vec{C}, \tag{7.5}$$

in the reciprocal space.

For any periodic function $f(\vec{r})$ in real space, it is natural to construct it using the discrete Fourier transform

$$f(\vec{r}) = \sum_{h,k,l=-\infty}^{\infty} C_{hkl}\mathrm{e}^{\mathrm{i}\vec{G}_{hkl}\cdot\vec{r}}. \tag{7.6}$$

Note that the phase factor $\vec{G}_{hkl}\cdot\vec{r}_{uvw}$ is

$$\vec{G}_{hkl}\cdot\vec{r}_{uvw} = (h\vec{A}+k\vec{B}+l\vec{C})\cdot(u\vec{a}+v\vec{b}+w\vec{c}) = 2\pi(hu+kv+wc), \tag{7.7}$$

which is always integer numbers of 2π.

Now we could see that the plane (hkl) is normal to the vector $\vec{G}_{hkl}$, and wave with wavevector $\vec{G}_{hkl}$ changes by 2π in phase between one plane and the next in the $\{hkl\}$ set.

For diffraction to happen, the periodic function is naturally $E(\vec{x}) = E_0\mathrm{e}^{\mathrm{i}(\vec{k}\cdot\vec{x}-\omega t)}$ we need to look at the incoming and outgoing wavevectors $\vec{k}_i$ and $\vec{k}_f$. Define the scattering wavevector $\vec{k}_s$ to be $\vec{k}_f - \vec{k}_i$.

When $\vec{k}_s$ is a reciprocal lattice vector, the extra phase[2] $\vec{k}_s\cdot\vec{r}$ is a multiple of 2π, as shown above. This will give a multiple of 2π phase factors, which adds all light in-phase. Hence, the condition for diffraction from the lattice is

$$\vec{k}_s = \vec{k}_f - \vec{k}_i = \vec{G}_{hkl}. \tag{7.8}$$

[1] Note that this is only for the cubic lattices. For general lattice we need to use the Weiss Zone Law.
[2] $\vec{r}$ is the vector from one scattering object to another.

This can be drawn in an *Ewald's sphere*, where $\vec{k}_i$ and $\vec{k}_f$ are two radius of the sphere pointing to different directions, and $\vec{G}$ connects the ends of them. Under some conditions, there may be no valid $\vec{G}$ that could link the wavevectors, then there is no diffraction from the lattice.

From IA materials we know that single crystal diffraction produces dots on the screen, and powder diffraction "spins" the dots to create rings on the screen, the angle can be measured using lots of ways and equipments.

7.3 From Normal Modes to Phonons

We want to study the motion of every atom in the crystal. Fortunately, the interaction is in the length scale where electromagnetic force dominates. Therefore, for every atom, the motion about its equilibrium position is (to first order) simple harmonic motion.

Indeed, the case is not that simple, since we're also working in the scale where quantum mechanics is not negligible. But let's work from the basics, starting from normal modes.

7.3.1 One-dimensional harnomic chain

We assume that we have $N \to \infty$ number of mass points, m, connected in series with springs that have spring constant k. Suppose the n-th mass point is x_n away from its equilibrium position, the equation of motion is simply

$$m\ddot{x}_n = k(x_{n-1} - 2x_n + x_{n+1}). \tag{7.9}$$

Conventionally, the equation can be solved by solving the equation

$$\begin{vmatrix} 2k - m\omega^2 & -k & 0 & \cdots & -k \\ -k & 2k - m\omega^2 & -k & \cdots & 0 \\ 0 & -k & 2k - m\omega^2 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ -k & 0 & 0 & \cdots & 2k - m\omega^2 \end{vmatrix}_{\infty\times\infty} = 0, \tag{7.10}$$

but this is clearly too complicated to write out and simplify. Therefore, we make some simplifications on the system.

- Cyclic boundary conditions: $x_{N+1} = x_1$.
- All atoms have the same amplitude u_0 and frequency ω, due to the symmetry that all atoms are equivalent.

Now it's safe to say that there can only be a constant phase shift δ between one atom and the next,

$$u_n = u_0 \mathrm{e}^{\mathrm{i}(n\delta - \omega t)}. \tag{7.11}$$

Therefore, our equation of motion becomes,

$$-m\omega^2 = k(\mathrm{e}^{\mathrm{i}\delta} + \mathrm{e}^{-\mathrm{i}\delta} - 2), \qquad \omega = \sqrt{\frac{4k}{m}} \left|\sin\left(\frac{\delta}{2}\right)\right|. \tag{7.12}$$

Note that, the frequency of oscillation is a **function** of δ, which means we have N possible frequencies. This is valid since we have N atoms and thus N order of magnitude. The function is a even, and is monotonic from 0 to π. Generally, since ω becomes periodic when the range of δ is larger than 2π, we usually just plot the function over the range $[-\pi, \pi]$. This range is called the *First Brillouin Zone* (BZ). If something goes out of the first BZ, we will subtract the δ by 2π and represent it again within the first BZ. This is valid according to translational symmetry.

Note that, the first BZ is the Wigner-Seitz unit cell in the reciprocal lattice, that's why it contains every information in the crystal.

Long- and short-wave limits Write the phase factor δ as ka, where $k = 2\pi/\lambda$ is the wavevector[3] and a is the distance between adjacent atoms.

When the wavelength is large ($q \to 0$), the wave is equivalent to be traveling in a continuous media. Therefore naturally, wavespeed tends to the speed of sound, $v = \sqrt{K_{\text{bulk}}/\rho}$.

The largest unique wavevector is of magnitude π/a, this is from the quantum property of phonons. The dispersion relationship becomes $\omega = \sqrt{4\alpha/m}$, which is a (effectively) standing wave since group velocity[4] $\partial\omega/\partial k = 0$.

Higher-order terms Although we claimed that the function $\omega(\delta)$ is even and monotonic, it is not always the case. Let's consider the most general case, where the force is EM but not simply elasticity. Now the function remains even because of symmetry, but is no longer necessarily monotonic, because longer range interactions are possible

$$m\ddot{x}_n = \cdots + k'x_{n-2} + kx_{n-1} - 2kx_n + kx_{n+1} + k'x_{n+2} + \cdots . \tag{7.13}$$

This gives,

$$\omega^2 = \frac{4k}{m}\left[\sin^2\left(\frac{\delta}{2}\right) + \frac{k'}{k}\sin^2(\delta) + \cdots\right]. \tag{7.14}$$

Even now the picture is still not complete. Real systems are not even harmonic because of many reasons, the main consequence is that phonons can interact with each other.

1D diatomic lattice For diatomic lattice in the form ABAB we need to change our equations of motion

$$\begin{aligned} m_A\ddot{x}_{2n} &= k(x_{2n-1} + x_{2n+1} - 2x_{2n}); \\ m_B\ddot{x}_{2n+1} &= k(x_{2n} + x_{2n+2} - 2x_{2n+1}). \end{aligned} \tag{7.15}$$

Another change to be made is, we can't assume equal amount of amplitudes now. Suppose[5]

$$x_{2n} = U_1 \mathrm{e}^{\mathrm{i}(2nqa - \omega t)}, \qquad x_{2n+1} = U_2 \mathrm{e}^{\mathrm{i}((2n+1)qa - \omega t)}, \tag{7.16}$$

[3] Do not confuse it with the spring constant k.
[4] Velocity of phonon.
[5] Don't forget the phase factor $\delta = nq$ for each particle.

we have

$$
\begin{aligned}
(m_A\omega^2 - 2k)U_1 + 2k\cos(qa)U_2 = 0;\\
2k\cos(qa)U_2 + (m_B\omega^2 - 2k)U_2 = 0.
\end{aligned}
\tag{7.17}
$$

Taking the determinant, for each n we get two possible values of ω^2,

$$
\omega^2 = \frac{k}{m_A m_B}\left\{(m_A + m_B) \pm \left[(m_A + m_B)^2 - 4m_A m_B \sin^2(qa)\right]^{1/2}\right\}, \tag{7.18}
$$

therefore it remains true that the number of frequencies are equal to the number of particles.

In the ω-q plot, the two solutions of ω form two branches in the dispersion relationship. The lower branch (neighbouring atoms oscillates in phase) is called the *acoustic (A) mode*, which corresponds to sound waves in the long wavelength limit. The upper branch (neighbouring atoms oscillates out of phase) is called the *optical (O) mode*, it interacts strongly with electromagnetic radiation in polar crystals, and leads to strong optical absorption, because it is of higher energy (frequency). In the limits of $q = \pm\pi/2a$ (standing waves) and $q \to 0$ (sound waves), there are fum properties to look at, but the picture is easy to imagine.

If we think of a monoatomic chain, it can be regarded as a diatomic chain with $m_A = m_B$. Therefore the first BZ is half its original size. Modes outside the new first BZ must be *backfolded* into the zone. We arrive at similar patterns to what we derived, but with $m_A = m_B$, the two branches are connected at $q = \pi/2a$. If the diatomic chain has different mass, the connection will break, and give us the two branches we wanted.

7.3.2 Phonons

We quantise phonons by the N number of modes on the harmonic chain. For N not necessarily approaching infinity, the allowed k values are discrete,

$$
k = \frac{2\pi n}{Na}, \tag{7.19}
$$

where each mode n is a quantised harmonic oscillator with energy

$$
E = \left(n + \frac{1}{2}\right)\hbar\omega. \tag{7.20}
$$

This - phonons - is probably the first quasi-particle we have ever met in physics, believe me, there are more to come.

Phonons have particle character, and obeys Bose-Einstein (BE) statistics. Its energy is quantised by $\hbar\omega$, momentum by $\hbar k$, and velocity $\partial\omega/\partial k$.

But remember that k depends on a, the spacing between lattice points. That means the momentum of phonons are actually determined by the crystal. Therefore, $\hbar k$ is also called the *crystal momentum*. That can be understood, however, by considering the movement of lattice: A phonon distorts the lattice so instantaneously, that the lattice now looks like a lattice with a longer repeat distance.

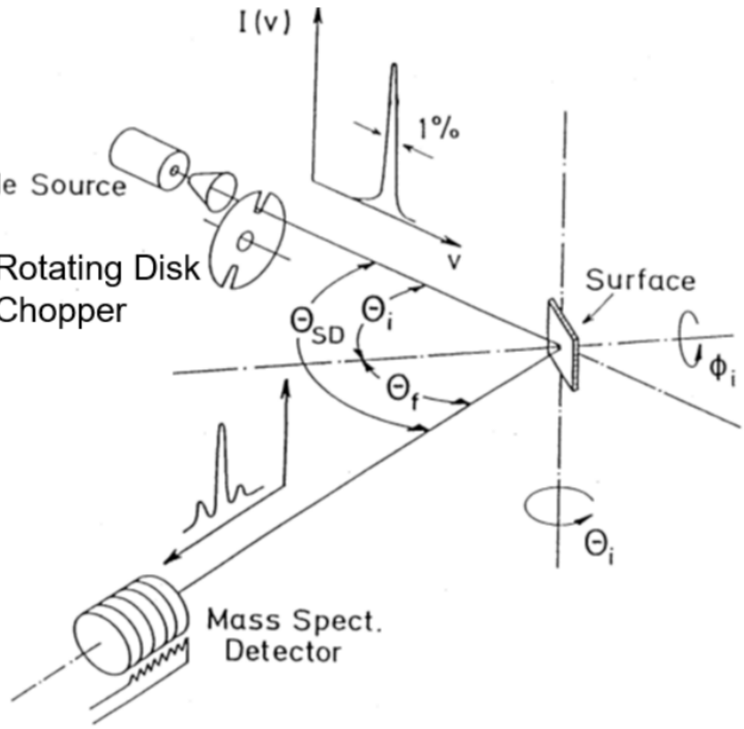

Figure 1. In the device, scattering angles can be varied to control the directions of $\vec{k}_i$ and $\vec{k}_f$. 'Time of Flight' is used to determine the energy transfer on scattering, $\hbar\omega$ [2].

If an external particle[6] comes in and diffracts through the lattice, the **scattering wavevector** $\vec{k}_s = \vec{k}_f - \vec{k}_i$ would be simply the **crystal momentum**, or the **phonon wavevector** $\vec{k} + \vec{G}$ (any $\vec{G}$ vector is possible because $\vec{k} + \vec{G}$ returns to the first BZ). This can also be understood as the external particle picks up momentum $\hbar(\vec{k} + \vec{G})$ from the phonon by annihilation of a phonon of wavevector $(\vec{k} + \vec{G})$, or creation[7] of a phonon with wavevector $-(\vec{k} + \vec{G})$.

Phonons can only be measured using creation and annihilation processes. This is often carried out by thermal energy **neutrons** for bulk phonons and thermal energy **He** atoms for surface phonons[8]. Then, find the initial and final wavevectors and apply conservation laws:

$$\hbar\omega = \frac{\hbar^2}{2m}\left(k_i^2 - k_f^2\right); \qquad \vec{k}_i = \vec{k}_f + \vec{k} + \vec{G} \quad \text{for phonon creation.} \tag{7.21}$$

In practice, we would have an apparatus shown in Fig. 1, examples include [3] for He atom scattering and [4, 5] for neutron scattering.

[6] Called *probe particle*. Often neutrons, because they have similar wavelengths.

[7] Because phonons are bosons, they are antiparticles of themselves. This can be investigated quantitatively by applying Klein-Gordon theory on crystals.

[8] X-rays have the correct wavelength but their energy is too high, up to keV scale, while phonons have energies of order 10 meV.

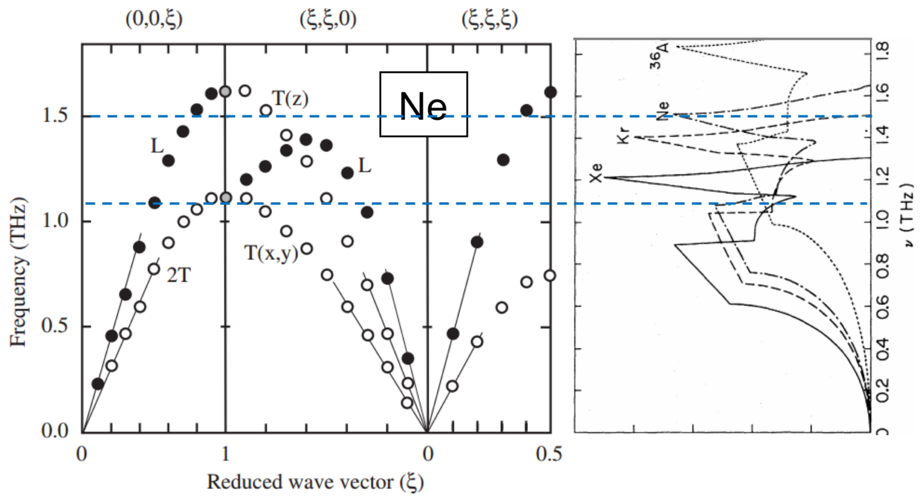

Figure 2. Phonon band structure of crystal Ne [6]. The LHS plot is just the longitudinal (L) and transverse (T) modes (each point is a mode obtained from diffraction). The RHS gives the density of states $g(\omega) = \mathrm{d}N/\mathrm{d}\omega$ as a function of ω. We could see that the peaks form a one-one relationship to the material.

7.3.3 Phonons in 3D Lattices

It is impossible to study phonons fully because in 3D there are infinite number of directions a wave can travel. But we can always look at the projections to the symmetric axes. For example, in FCC these directions are (100), (110) and (111), we will see plots like Fig. 2. From the plot we can see that transverse modes are often degenerate along high symmetry directions, but not along directions like ($\xi\xi0$). Also, the L mode along ($\xi\xi0$) contains two Fourier components, suggesting higher order terms. That's because we need to consider the full structure, an atom may have nearest neighbours in different planes[9], but not really the effect of longer range interactions.

Things gets more complicated in diatomic lattices, as in each range we will have at least four modes, TO, LO, TA and LA. We put "at least" there because the transverse modes in the ($\xi\xi0$) directions are not degenerate. An example is NaCl [7], as shown in Fig. 3. Note that, modes with the same symmetry[10] cannot cross, for example, the LO and LA modes cannot cross in the ($\xi\xi0$) and ($\xi00$) directions.

Phonon band structure is a basic property to be tested for a given material, which can be experimentally measured by elastic neutron scattering. Others include X-ray diffraction for lattice structure, various ways of determining bulk modulus, etc. For a typical procedure of phonon bands investigation, see [8, 9].

[9] For example, in FCC both $(1, 0, 0)$ and $(1/2, 1/2, 0)$ are the nearest neighbours of the atom at $(0, 0, 0)$, but they are at different distances to the origin.

[10] This is the geometrical symmetry, such as reflection, translation, rotation, etc. Not the specific modes.

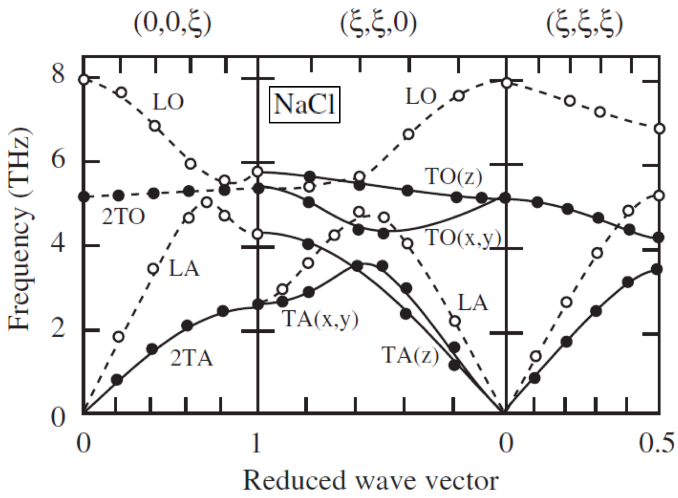

Figure 3. Phonon band structure of NaCl, comparing with that of Ne, one may observe that the highest optical frequencies are much higher here in NaCl. That is because bonds in NaCl are polar bonds, which are much stronger (more bond energy) than Van der Waals bonding in Ne.

7.4 Thermal Properties of Insulators

For a given material, the most important thermal properties are probably heat capacity and thermal conductivity. In 19th century, there are already some observations into those properties. In 1819, Dulong and Petit [10] found that the heat capacity per atomic weight of a substance is approximately constant ($3k_{\mathrm{B}}$ per atom) near room temperature. Other observation state that heat capacity of insulators[11] usually varies proportionally to T^3 at low temperatures.

But why are thermal properties related to phonons? Recall that temperature is a reflection of kinetic energy of atoms. Since phonons are a way to describe the motion inside a crystal, it should naturally be related to the thermodynamics of the crystal. Now we use the Boltzmann distribution to calculate its internal energy and see if we can find something out.

Ignoring the ground-state energy $\hbar\omega/2$, the average energy E_i in the i-th mode is obviously

$$E_i = \frac{\sum_{n=0}^{\infty} n\hbar\omega_i \exp\left(-\frac{n\hbar\omega}{k_{\mathrm{B}}T}\right)}{\sum_{n=0}^{\infty} \exp\left(-\frac{n\hbar\omega}{k_{\mathrm{B}}T}\right)}, \tag{7.22}$$

let $\beta = 1/k_{\mathrm{B}}T$, we have

$$E_i = \frac{\hbar\omega_i}{\exp(\hbar\omega_i\beta) - 1}. \tag{7.23}$$

Does it solves the problem, i.e., does this describes the phenomena above? Yes. At low temperatures, $E_i \simeq \hbar\omega_i \exp(-\hbar\omega_i\beta)$, at high temperatures, $E_i \simeq k_{\mathrm{B}}T$. If we have N modes, the heat capacity will be precisely $\partial(3NK_{\mathrm{B}}T)/\partial T = 3Nk_{\mathrm{B}}$.

[11] Why insulators? Because electrons plays a significant role in materials, even in thermal properties!

Normally, in a material we have several Avogadro constant's number of atoms. Therefore, we can convert the sum into an integral for the total internal energy U:

$$U = \int_0^\infty \frac{\hbar\omega}{\exp(\hbar\omega\beta) - 1} g(\omega)\ \mathrm{d}\omega. \tag{7.24}$$

where $g(\omega) = \frac{\mathrm{d}N}{\mathrm{d}\omega}$ is the density of states which describes the energy distribution of phonons[12]. Until now we have used nothing else than thermodynamics, the problem is we haven't found $g(\omega)$, the density of states. Here's where Debye model comes over and help us out.

7.4.1 Debye model for heat capacity

Debye model was published in 1912 so I don't know if Peter Debye borrowed any thought from the Rayleigh-Jeans Law, but they do share the same idea: play with waves in a box and consider the allowed modes in the first octant of k-space. Consider a box of sides A, B and C, the number of states in the shell from k to $k + \mathrm{d}k$ is[13]

$$\mathrm{d}N = g(k)\ \mathrm{d}k = \frac{3\frac{4\pi k^2}{8}\ \mathrm{d}k}{\frac{\pi^3}{ABC}}, \tag{7.25}$$

therefore

$$g(k) = \frac{3Vk^2}{2\pi^2} \qquad \Rightarrow \qquad g(\omega) = g(k)\frac{\mathrm{d}k}{\mathrm{d}\omega}. \tag{7.26}$$

As long as we know the phonon dispersion relationship, we can obtain $g(\omega)$. Note that, the theory assumed perfect reflecting boundary conditions as we did in blackbody radiation, therefore phonons form standing waves in the crystal. Later when we talk about electrons, we use the cyclic boundary conditions, which will lead to travelling waves and different statistics.

Debye then did the long-wavelength approximation that $\omega = v_s k$, which "ignores the atomic nature of the material [2]". But I don't like this expression, it's just about working at another length scale. Following this dispersion relations, it is easy to get

$$g(\omega) = \frac{3V\omega^2}{2\pi^2 v_s^2} \propto \omega^2. \tag{7.27}$$

The quadratic proportionality cause a problem with normalisation, since the density of states should normalise to the number of states. Therefore we need to stop integrating at a particular *Debye frequency* ω_D, which is defined by

$$3N = \int_0^{\omega_D} g(\omega)\ \mathrm{d}\omega \qquad \Rightarrow \qquad \omega_D^3 = \frac{6\pi^2 v_s^3 N}{V}. \tag{7.28}$$

[12] Phonons are bosons, their energy states can be degenerate, therefore we don't need more complicated statistics.

[13] We included the factor if 3 to allow for 2 transverse modes and 1 longitudinal mode.

The internal energy and heat capacity are therefore

$$
\begin{aligned}
U &= \frac{3V\hbar}{2\pi^2 v_s^2} \int_0^{\omega_D} \frac{\omega^3}{\exp(\hbar\omega\beta) - 1}\,\mathrm{d}\omega; \\
C &= \frac{\partial U}{\partial T} = 9Nk_\mathrm{B} \left(\frac{T}{\theta_D}\right)^3 \int_0^{\theta_D/T} \frac{x^4 \mathrm{e}^x}{(\mathrm{e}^x - 1)^2}\,\mathrm{d}x.
\end{aligned}
\tag{7.29}
$$

Here θ_D is the Debye temperature, which is given by $\hbar\omega_D = k_\mathrm{B}\theta_D$. The heat capacity rises raidly at first then saturates towards $3Nk_\mathrm{B}$ because the integrating factor can be approximated as x^2 at high temperatures. At low temperatures, we obtain the T^3 law because we can simply integrate to infinity[14]. To sum up,

$$
C(T \to \infty) = 3Nk_\mathrm{B}; \qquad C(T \to 0) = \frac{12\pi^4}{5} Nk_\mathrm{B} \left(\frac{T}{\theta_D}\right)^3. \tag{7.30}
$$

We can already see from the RHS of Fig. 2 that the measured $g(\omega)$ in crystal gas is quite different from a quadratic function, in metals similar things happen [11]. But luckily Dobye model does fit well in the low energy (ω) region, the difference was caused by (i) the 3D first BZ has a more complicated shape and (ii) both transverse and longitudinal modes have different dispersion curves.

More on Debye frequencies and temperatures The Debye frequency is a characteristic frequency controlled by the **mass** of atoms and **stiffness** of lattice, this can be seen in Eq. 7.28. As we move down the periodic table, atomic mass increases and atoms become more deformable, θ_D decreases as expected. For transition metals, θ_D is generally between 200 and 600 K, which explains the roughly constant heat capacity at room temperature. But the θ_D for carbon is 2230 K, this relates to the unusual properties of graphite, graphene and diamond.

Note that, in exams the term "heat capacity" can be any one of:

- The total heat capacity of the sample (extensive).
- Heat capacity per atom (intensive).
- Heat capacity per unit mass (intensive), or specific heat.
- Heat capacity per unit volume (intensive), or volumetric heat capacity.
- Heat capacity per mole (intensive), or molar heat capacity.

The examiners, as they always do, expect students to understand which heat capacity they are referring to from context. The clever way to do that is use the information given and perform dimensional analysis. As exam questions do always have answers, the information given must indicate correct units of the "heat capacity" used.

[14] The integral is purely numerical and gives $4\pi^4/15$.

7.4.2 Thermodynamical treatment for thermal conductivity

For thermal conductivity we need to define an energy carrier, which is obviously phonon. Suppose a phonon travels with velocity v and velocity distribution $f(v)$ for a certain distance $\bar{l}$ before interacting and transferring energy[15], and crosses the isothermal plane at an angle θ. The excess temperature and energy of phonons crossing the plane would be

$$\Delta T = \frac{\mathrm{d}T}{\mathrm{d}z}\Delta z = -\frac{\mathrm{d}T}{\mathrm{d}z}\bar{l}\cos\theta; \qquad \Delta T = c_{ph}\Delta T = -c_{ph}\frac{\mathrm{d}T}{\mathrm{d}z}\bar{l}\cos\theta, \tag{7.31}$$

where c_{ph} is the heat capacity of a phonon mode. The total heat flow through the plane will be the integral

$$\begin{aligned}\frac{\mathrm{d}^2 Q}{\mathrm{d}S\mathrm{d}t} &= \int_{v=0}^{\infty}\int_{\theta=0}^{\pi}[nvf(v)\cos\theta\ \mathrm{d}v]\cdot\left[-c_{ph}\frac{\mathrm{d}T}{\mathrm{d}z}\bar{l}\cos\theta\cdot\frac{1}{2}\sin\theta\ \mathrm{d}\theta\right]\\ &= -\frac{1}{2}c_{ph}n\bar{l}\frac{\mathrm{d}T}{\mathrm{d}z}\int_0^{\infty} vf(v)\ \mathrm{d}v\int_0^{\pi}\sin\theta\cos^2\theta\ \mathrm{d}\theta\\ &= -\frac{1}{3}c_{ph}n\bar{l}\bar{v}\frac{\mathrm{d}T}{\mathrm{d}z} = -\kappa\frac{\mathrm{d}T}{\mathrm{d}z},\end{aligned} \tag{7.32}$$

where $\kappa = \frac{1}{3}c_{ph}n\bar{l}\bar{c} = \frac{1}{3}C\bar{l}\bar{c}$ and C is the heat capacity per unit volume. Note that this is identical to the thermal conducting law in gas and liquids. The only change was the heat carrier.

But what are the mean free path and mean velocity of phonons? Mean free path is reduced by scattering processes inside the crystal, and the relation is

$$\Gamma = \frac{\bar{c}}{\bar{v}}, \tag{7.33}$$

where Γ is the total scattering rate, and $1/\bar{l}$ should be the sum of inverse of free paths. There are several kinds of scatterings, such as the *geometric scattering* from sample or grain boundaries, and *phonon-phonon (ph-ph) scattering* in an anharmonic lattice.

In pure crystalline form, κ for insulators such as diamonds are generally much higher than metals, but non-crystalline systems such as glass have much lower values of κ.

At low temperatures, thermal conductivity is usually proportional to T^3 [12, 13], and T^{-1} at high temperatures [14]. That is because, at low temperatures there are fewer phonons, so ph-ph scatterings are insignificant, $C \propto T^3$ following Debye model. But at high temperatures more phonons are excited, ph-ph scattering now dominates. Since C has saturated to constant at these temperatures, $\kappa \propto \bar{l} \propto 1/N \propto 1/T$.

But even for ph-ph scatterings there are different processes. Generally, it obeys energy and momentum conservation[16], but since we have the backfolding for wavevectors outside the first BZ, there are:

- Normal (N) scattering: Don't change the resulting wavevector dramatically (simple addition), therefore weak effect on thermal conductivity.

[15] $\bar{l}$ is the mean free path of phonons.

[16] Still remember that phonons are bosons, and creation of phonon can be viewed as annihilation of another phonon.

- Umklapp (U) scattering: The resulting wavevector is outside the first BZ, therefore changes the wavevector dramatically when folding back into the first BZ. But the U-processes only dominates at very high temperatures because they need relatively high energy and momentum.

Now we could explain why the thermal conductivity goes above the $1/T$ relation at intermediate temperatures.

7.5 The Free Electron Model

7.5.1 Density of states for free electrons

Now we shall move on to metals and find the density of states $g(\omega)$ in them.

In conductors, especially metals, phonons do not play a significant role in carrying energy anymore, electrons take the responsibility. Therefore, the density of states we are looking at should be the density of states for electrons. And here are some assumptions before everything.

- Only consider valance electrons.
- Electrons can be treated classically, so they are balls in a pinball machine (lattice).

These assumptions were originally made by Paul Drude in 1900, about the same time with Rayleigh-Jeans law, so we can see how much change does the "sphere in k space" had brought to physics in such a short time. The model is now called the *classical Drude model*, and worked remarkably well in describing electrical conductivity, optical reflectivity and thermal conductivity, etc. The *free electron model* combines it with the Fermi-Dirac statistics for electrons, and made some further assumptions

- Electrons are free to move through the lattice as quantum waves. And are only affected by (cyclic) boundary conditions.
- Ignore electron-electron repulsion.
- The Pauli exclusion principle applies, i.e. a quantum state can support a maximum of two electrons with opposite spins.
- Positive charge from the ion cores forms a *uniform* background to maintain charge neutrality.

The last assumption is probably the most important because it can lead to dramatic change in effective mass m^*, we will discuss this later.

Now write the assumptions in mathematical language. The states form a lattice of points in k space following the cyclic boundary conditions, each state occupies a volume of $(2\pi)^3/V$. But we need to consider the full space rather than the first octant, because k can be both positive and negative.

$$\mathrm{d}N = g(k)\ \mathrm{d}k = 2\frac{4\pi k^2\ \mathrm{d}k}{\frac{(2\pi)^3}{V}} = \frac{Vk^2}{\pi^2}\ \mathrm{d}k. \tag{7.34}$$

Since $\epsilon = \frac{\hbar^2 k^2}{2m}$, we have

$$g(\epsilon) = \frac{\mathrm{d}N}{\mathrm{d}\epsilon} = g(k)\frac{\mathrm{d}k}{\mathrm{d}\epsilon} = \frac{Vkm}{\hbar^2\pi^2} = \frac{V}{2\pi^2}\left(\frac{2m}{\hbar^2}\right)^{3/2}\epsilon^{1/2} \propto \epsilon^{1/2}. \tag{7.35}$$

Note that this is only valid for three-dimensional materials, in a random integer dimension n,

$$\mathrm{d}N = g(k)\ \mathrm{d}k = 2\frac{\frac{2\pi^{n/2}}{\Gamma(n/2)}k^{n-1}\ \mathrm{d}k}{\frac{(2\pi)^n}{V_n}} = \frac{4V_n k^{n-1}}{(4\pi)^{n/2}\Gamma(n/2)}\ \mathrm{d}k, \tag{7.36}$$

therefore,

$$g(\epsilon) = \frac{4V_n k^{n-2}m}{(4\pi)^{n/2}\hbar^2\Gamma(n/2)} = \frac{2V_n m^{n/2}\epsilon^{\frac{n}{2}-1}}{(2\pi)^{n/2}\hbar^n\Gamma(n/2)} \propto \epsilon^{\frac{n-2}{2}}. \tag{7.37}$$

For example, in two dimensions $g(\epsilon)$ is proportional to ϵ^0, therefore constant; in one dimension $g(\epsilon)$ is proportional to $\epsilon^{-1/2}$, as derived.

7.5.2 Fermi-Dirac distribution

At positive and finite temperature T, electrons obey *Fermi-Dirac (FD) distribution*

$$p(\epsilon) = \frac{1}{\exp\left(\frac{\epsilon-\mu}{k_\mathrm{B}T}\right)+1}. \tag{7.38}$$

There are several ways to derive it [15], we will introduce the thermal dynamical approach instead of the statistical one[17].

The first law of thermodynamics,

$$\mathrm{d}U = T\mathrm{d}S - P\mathrm{d}V + \mu\mathrm{d}N, \qquad \mu = \left(\frac{\partial U}{\partial N}\right)_{S,V}. \tag{7.39}$$

At constant V, consider electron state with energy ϵ with a heat and particle reservoir. In the ground state (electron state unoccupied), the entropy of the reservoir is $S_0 = k_\mathrm{B}\ln(\Omega_0)$, where Ω_0 is the number of reservoir configurations. To occupy the state at energy ϵ, an electron of energy ϵ must be trasferred from the reservoir, which leads to an entropy raise,

$$S_0 + \mathrm{d}S = S_0 - \frac{\epsilon}{T} + \frac{\mu}{T} = k_\mathrm{B}\ln\Omega, \qquad \mathrm{d}\ln\Omega = -\frac{\epsilon-\mu}{k_\mathrm{B}T}. \tag{7.40}$$

Therefore the property of the state being occupied is

$$p(\epsilon) = \frac{0\times 1 + \exp[-(\epsilon-\mu)/k_\mathrm{B}T]\times 1}{1+\exp[-(\epsilon-\mu)/k_\mathrm{B}T]} = \text{FD distribution}. \tag{7.41}$$

At $T = 0$, this is simply a step function which ends at $\epsilon = \mu$. But at finite non-zero T, the end of the distribution get "smeared out" by $k_\mathrm{B}T$ by thermal excitations. This gave rise to the exponential tail in Millikan's experiment measuring Planck's constant h [16, 17].

From the FD distribution, it is clear that the chemical potential (Gibbs free energy per mole) is the energy at which a state has 50% chance of being occupied. The value of

[17] Personally, I prefer the statistical approach as it is more straightforward.

$\mu(T)$ at $T = 0$ is called the *Fermi energy*. The value of it should be given under desired conditions, otherwise there's no way to figure out the exact distribution. Luckily, chemists have worked out a tables for this.

From numerical analysis, Fermi energy is approximately the chemical potential at low temperatures.

7.5.3 Electronic heat capacity

From the density of states function and Fermi-Dirac distribution it is clear that electrons fill up to approximately the chemical potential, except smearing out by a standard width $k_{\mathrm{B}}T$. The internal energy is easily calculated

$$U = \int_0^\infty \epsilon g(\epsilon) p(\epsilon)\ \mathrm{d}\epsilon, \tag{7.42}$$

where we don't afraid integrating to infinity because normalisation is preserved by the FD distribution function.

Now we must apply a trick called *Sommerfeld expansion* [18], this is to deal with integrals containing the distribution function, which is in our case the density of states. For a integral of the form

$$\int_0^\infty f(\epsilon) h(\epsilon)\ \mathrm{d}\epsilon,$$

we define $Q(\epsilon)$ s.t. $h(\epsilon) = \mathrm{d}Q(\epsilon)/\mathrm{d}\epsilon$, or

$$Q(\epsilon) = \int_0^\epsilon h(\epsilon')\ \mathrm{d}\epsilon'.$$

Now we have

$$\int_0^\infty f(\epsilon) h(\epsilon)\ \mathrm{d}\epsilon = \int_0^\infty f(\epsilon)\ \mathrm{d}Q(\epsilon) = f(\epsilon)Q(\epsilon)|_0^\infty + \int_0^\infty Q(\epsilon)\left(-\frac{\partial f(\epsilon)}{\partial \epsilon}\right)\ \mathrm{d}\epsilon.$$

Since $Q(0) = 0$ and $f(\infty) = 0$,

$$\begin{gathered} \int_0^\infty f(\epsilon) h(\epsilon)\ \mathrm{d}\epsilon = \int_0^\infty Q(\epsilon)\left(-\frac{\partial f(\epsilon)}{\partial \epsilon}\right)\ \mathrm{d}\epsilon; \\ -\frac{\partial f}{\partial \epsilon} = \frac{1}{k_{\mathrm{B}}T\left(\mathrm{e}^{x/2} + \mathrm{e}^{-x/2}\right)^2}, \quad x = \frac{\epsilon - \mu}{k_{\mathrm{B}}T}. \end{gathered} \tag{7.43}$$

This is a normalised peak at $\epsilon = \mu$ with mirror symmetry about $x = \mu$ and width $\sim k_{\mathrm{B}}T$. At low temperatures[18], the peak is very narrow, which allows us to write it as $\delta(\epsilon - \mu)$ and integrate from $-\infty$ to ∞. At the same time, we shall expand $Q(\epsilon)$ near $\epsilon = \mu$

$$Q(\epsilon) = Q(\mu) + Q'(\mu)(\epsilon - \mu) + \frac{1}{2}Q''(\mu)(\epsilon - \mu)^2 + \mathcal{O}\left((\epsilon - \mu)^3\right),$$

[18] This can be a definition of chemical potential μ for fermions: μ is whatever energy that makes the FD distribution normalised.

and plug everything into the integral above.

$$
\begin{aligned}
\int_0^\infty f(\epsilon)h(\epsilon)\ \mathrm{d}\epsilon &= \int_0^\infty Q(\epsilon)\left(-\frac{\partial f(\epsilon)}{\partial \epsilon}\right)\ \mathrm{d}\epsilon \\
&= Q(\mu)\int_{-\infty}^{\infty}\left(-\frac{\partial f(\epsilon)}{\partial \epsilon}\right)\ \mathrm{d}\epsilon + Q'(\mu)\int_{-\infty}^{\infty}(\epsilon-\mu)\left(-\frac{\partial f(\epsilon)}{\partial \epsilon}\right)\ \mathrm{d}\epsilon \\
&\quad + \frac{Q''(\mu)}{2}\int_{-\infty}^{\infty}(\epsilon-\mu)^2\left(-\frac{\partial f(\epsilon)}{\partial \epsilon}\right)\ \mathrm{d}\epsilon \\
&= Q(\mu)\int_{-\infty}^{\infty}\delta(\epsilon-\mu)\ \mathrm{d}\epsilon + Q'(\mu)\int_{-\infty}^{\infty}(\epsilon-\mu)\delta(\epsilon-\mu)\ \mathrm{d}\epsilon \\
&\quad + \frac{Q''(\mu)}{2}\int_{-\infty}^{\infty}(\epsilon-\mu)^2\left(-\frac{\partial f(\epsilon)}{\partial \epsilon}\right)\ \mathrm{d}\epsilon \\
&= Q(\mu) + 0 + \frac{Q''(\mu)}{2}\int_{-\infty}^{\infty}(\epsilon-\mu)^2\left(-\frac{\partial f(\epsilon)}{\partial \epsilon}\right)\ \mathrm{d}\epsilon \\
&= Q(\mu) + \frac{Q''(\mu)}{2}(k_\mathrm{B}T)^2\int_{-\infty}^{\infty}\frac{x^2}{\left(\mathrm{e}^{x/2}+\mathrm{e}^{-x/2}\right)^2}\ \mathrm{d}x \\
&= Q(\mu) + \frac{\pi^2}{6}Q''(\mu)(k_\mathrm{B}T)^2
\end{aligned}
$$

To get the internal energy, we need to first find the chemical potential, where $Q(\epsilon) = \int_0^\epsilon g(\epsilon')\ \mathrm{d}\epsilon'$. The number of electrons N is given by, at any small T,

$$
N = Q(\mu) + \frac{\pi^2}{6}Q''(\mu)(k_\mathrm{B}T)^2, \tag{7.44}
$$

which is our normalisation constant, Therefore[19],

$$
0 = Q'(\mu)\frac{\partial \mu}{\partial T} + \frac{\pi^2}{6}Q'''(\mu)\frac{\partial \mu}{\partial T}(k_\mathrm{B}T)^2 + \frac{\pi^2}{3}Q''(\mu)k_\mathrm{B}^2T.
$$

Differentiate again,

$$
\begin{aligned}
0 &= Q''(\mu)\left(\frac{\partial \mu}{\partial T}\right)^2 + Q'(\mu)\frac{\partial^2 \mu}{\partial T^2} \\
&\quad + \frac{\pi^2}{6}Q''''(\mu)\left(\frac{\partial \mu}{\partial T}\right)^2(k_\mathrm{B}T)^2 + \frac{\pi^2}{6}Q'''(\mu)\frac{\partial^2 \mu}{\partial T^2}(k_\mathrm{B}T)^2 + \frac{\pi^2}{6}Q'''(\mu)\frac{\partial \mu}{\partial T}k_\mathrm{B}^2T \\
&\quad + \frac{\pi^2}{3}Q'''(\mu)\frac{\partial \mu}{\partial T}k_\mathrm{B}^2T + \frac{\pi^2}{3}Q''(\mu)k_\mathrm{B}^2.
\end{aligned}
$$

This looks complicated, but if we take $T = 0$, it goes short,

$$
0 = Q'(\mu)\frac{\partial^2 \mu}{\partial T^2} + \frac{\pi^2}{3}Q''(\mu)k_\mathrm{B}^2, \qquad T = 0.
$$

Therefore the chemical potential near $T = 0$ is, written in Taylor expansion,

$$
\mu(T \to 0) = \epsilon_f - \frac{\pi^2}{6}\frac{g'(\epsilon_f)}{g(\epsilon_f)}(k_\mathrm{B}T)^2 = \epsilon_f\left[1 - \frac{\pi^2}{12}\left(\frac{k_\mathrm{B}T}{\epsilon_f}\right)^2\right]. \tag{7.45}
$$

[19]Take $T = 0$ for the equation below, we can find $\frac{\partial \mu}{\partial T}$ is zero at $T = 0$.

Now we move on and consider the internal energy, where $h(\epsilon) = \epsilon g(\epsilon)$. We assume $k_{\mathrm{B}}T \ll \mu$, therefore

$$\begin{aligned} Q(\mu) &= Q(\epsilon_f) + (\mu - \epsilon_f)Q'(\epsilon_f) \\ &= Q(\epsilon_f) - \frac{\pi^2}{6}\epsilon_f g'(\epsilon_f)(k_{\mathrm{B}}T)^2, \end{aligned}$$

which allows us to extract $Q''(\mu) = (\mu g'(\mu))'$. Plug in both expansions of $Q(\mu)$ and $Q''(\epsilon_f)$, we have

$$\begin{aligned} U &= Q(\mu) + \frac{\pi^2}{6}Q''(\mu)(k_{\mathrm{B}}T)^2 \\ &= Q(\epsilon_f) - \frac{\pi^2}{6}\epsilon_f g'(\epsilon_f)(k_{\mathrm{B}}T)^2 + \frac{\pi^2}{6}\epsilon_f g'(\epsilon_f)(k_{\mathrm{B}}T)^2 + \frac{\pi^2}{6}g(\epsilon_f)(k_{\mathrm{B}}T)^2 \\ &= Q(\epsilon_f) + \frac{\pi^2}{6}g(\epsilon_f)(k_{\mathrm{B}}T)^2, \end{aligned} \tag{7.46}$$

where $Q(\epsilon_f)$ is a constant. Therefore clearly,

$$C_V(\epsilon_f) = \frac{\partial U}{\partial T} = \frac{\pi^2}{3}g(\epsilon_f)k_{\mathrm{B}}^2 T, \tag{7.47}$$

Averaging across energy distribution, since approximation

$$g(\epsilon) \propto \epsilon^{1/2} \quad \Rightarrow \quad N = \int_0^{\epsilon_f} g(\epsilon)\ \mathrm{d}\epsilon = \frac{2}{3}\epsilon_f g(\epsilon_f)$$

holds at low temperature[20], we have

$$C_{V,\text{tot}} = \frac{\pi^2}{2}N\frac{k_{\mathrm{B}}^2 T}{\epsilon_f} = \frac{\pi^2}{2}Nk_{\mathrm{B}}\frac{T}{T_f} \propto T. \tag{7.48}$$

$T_f = \epsilon_f / k_{\mathrm{B}}$ is called the *Fermi temperature*, and T/T_f is typically 0.01 for metals at room temperature.

For the total heat capacity of the metal, we need to combine electrons and phonons. By the proportionality rules, it gives

$$C_V = \gamma T + \beta T^3 + \mathcal{O}(T^5). \tag{7.49}$$

The higher order terms, starting from fifth order, is generally small at low temperatures. Often we plot C/T against T^2 to obtain a straight line for data analysis [19].

Effective mass The effective mass m^* in free electron model is defined by $C_{\text{electrons}} \propto NTmn^{-2/3}$. Normally, electron-phonon coupling will increase m^*; strong electron-electron correlations can give huge values (thousands) of m^*/m. But interaction with the periodic potential from the lattice may cause $m^* < m$, which is abnormal, but does happen in Zn and Cd.

[20] This is just ignoring the smearing effect of FD distribution.

7.5.4 Electron pressure and isothermal bulk modulus

The free electron model, which treat electrons as waves & gas in a box, must lead to an outward pressure at the surface of the material. That is easy to picture in mind, because once you press the box and make it deform, the wavelengths shorten and the kinetic energies rise.

Since $\mathrm{d}U = T\mathrm{d}S - p\mathrm{d}V$, we will define p at low temperature as

$$p = -\left(\frac{\partial U}{\partial V}\right)_{T=0}. \tag{7.50}$$

We are now at absolute zero, so there is no need to consider FD distribution, we will assume that electrons fill up to the Fermi energy. Consider crystals in 3D, the average energy per electron is therefore,

$$\bar{\epsilon} = \frac{\int_0^{\epsilon_f} \epsilon g(\epsilon)\ \mathrm{d}\epsilon}{\int_0^{\epsilon_f} g(\epsilon)\ \mathrm{d}\epsilon} = \frac{\int_0^{\epsilon_f} \epsilon^{3/2}\ \mathrm{d}\epsilon}{\int_0^{\epsilon_f} \epsilon^{1/2}\ \mathrm{d}\epsilon} = \frac{3}{5}\epsilon_f. \tag{7.51}$$

Since we already have $N = \frac{2}{3}\epsilon_f g(\epsilon_f)$ and $\epsilon_f \propto (N/V)^{2/3}$,

$$p = -\frac{\partial U}{\partial V} = -\frac{\partial (N\bar{\epsilon})}{\partial \epsilon_f}\frac{\partial \epsilon_f}{\partial V} = -\left(\frac{3}{5}N\right)\left(-\frac{2}{3}\frac{\epsilon_f}{V}\right) = \frac{2}{5}n\epsilon_f. \tag{7.52}$$

The isothermal bulk modulus K_T is defined by

$$K_T = -V\left(\frac{\partial p}{\partial V}\right)_T. \tag{7.53}$$

Using the electron gas pressure derived above,

$$K_T = -V\left[\frac{\partial}{\partial V}\left(\frac{2}{5}\frac{N}{V}\epsilon_f\right)\right]_T = -V\left(\frac{2}{3}\frac{N}{V^2}\epsilon_f\right) = \frac{2}{3}n\epsilon_f. \tag{7.54}$$

As expected, the electron gas model contributes to only short-range repulsive interaction. It cannot compare to Coulombic attraction at longer range.

7.5.5 Properties under electromagnetic fields

Electrons are not moving completely freely inside the metal, it will collide with other particles, and give a mean drift **velocity** $\bar{v}$. This is characterised by the mean time between collisions τ,

$$\bar{v} = 0 \times [1 - \exp(t/\tau)] + v \times \exp(t/\tau) = v\exp(t/\tau),$$

therefore the drag acting on electrons is

$$\frac{\mathrm{d}\bar{v}}{\mathrm{d}t} = \frac{\bar{v}}{\tau},$$

and the equation of motion for **drift velocity** becomes

$$m^*\left(\frac{\mathrm{d}\vec{v}}{\mathrm{d}t} + \frac{\vec{v}}{\tau}\right) = -e(\vec{E} - \vec{v}\times\vec{B}). \tag{7.55}$$

This is the basic structure for all properties we will discuss below.

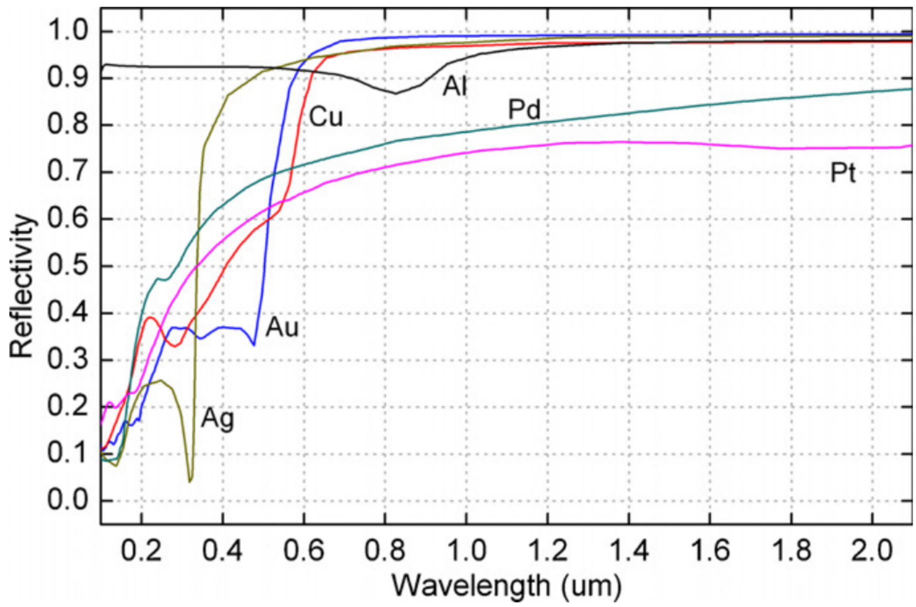

Figure 4. Reflectivity of some common metals versus wavelength at normal incidence [20, 21].

Electrical conductivity For electrons powered by a electric field along, the current density is simply

$$\vec{j} = -ne\vec{v} = \frac{ne^2\tau}{m^*}\vec{E} = ne\mu\vec{E} = \sigma\vec{E}, \quad \Rightarrow \quad \sigma = ne\mu, \tag{7.56}$$

where $\mu = v/E$ is called the electron mobility, and is a material-dependent quantity.

Since σ is generally high for metal, we treat free electrons in condensed matter in the same way as in a plasma, where the refractive index is given by

$$n_r = \sqrt{\epsilon_r} = \sqrt{1 - \frac{ne^2}{\epsilon_0 m^* \omega^2}} = \sqrt{1 - \frac{\omega_p^2}{\omega^2}}, \tag{7.57}$$

where $\omega_p^2 = ne^2/\epsilon_0 m^*$ is called the *plasma frequency*. This is covered in detail in the electromagnetism course.

At low frequencies, n is imaginary, giving high reflectivity and evanescent propagation into the metal; at high frequencies, n is real and the material is transparent. This may explain why metals has various colours in the visible region. For most metals, n is high therefore ω_p is in the UV range. But for transparent metals, ω_p can drop to or near IR. These transparent metals are widely used in the screen of smart phones and similar products because of their high electric conductivity.

But what exactly does the electric field does to electrons? We need to look at the Fermi sphere to consider all energy and momentum states.

Apply the quantum mechanical relation $\vec{p} = \hbar\vec{k} = m^*\vec{v}$, the equation of motion becomes

$$\hbar\left(\frac{\mathrm{d}\vec{k}}{\mathrm{d}t} + \frac{\vec{k}}{\tau}\right) - e\left(\vec{E} - \vec{v} \times \vec{B}\right). \tag{7.58}$$

Without collisions ($\tau \to \infty$), an electric field just causes the Fermi sphere to move steadily along the direction $-\vec{E}$ in $\vec{k}$ space. But with scattering, things are much different and will lead to many other properties for metals.

Electron scattering processes There are many kinds of scattering that can happen in metals, but here are the dominating two:

- Phonon scattering: Phonons have a wavevector comparable to electrons, but their energies[21] are tiny compared to ϵ_f. As a result, phonon scattering can **strongly change the direction** of an electron wavevector, but can only **slightly change the magnitude**.

- Defect scattering: Defects can also produce large changes in direction of $\vec{k}$, but only weak changes in energy.

These effects will add up using the reciprocal rule

$$\frac{1}{\tau} = \frac{1}{\tau_{\text{phonons}}} + \frac{1}{\tau_{\text{defects}}}. \tag{7.59}$$

At high temperature phonon scattering dominates. All phonon states are populated, and the number of phonon states is proportional to temperature. That means resistivity is proportional to temperature. At low temperatures, defect scattering dominates. Since defect is a sample (not even material) dependent quantity, resistivity will be a constant and temperature-independent offset, which is known as *Matthiessens's rule* [22]. These merely describes a general picture, and was experimentally tested by some of the best low-temperature physicists. The best fit if of course alkali metals [23], but for others at low temperature there are slight deviations [24].

Recall that the scattering wavevector $\vec{k}_s = \vec{k}_f - \vec{k}_i$, we need a vector in $\vec{k}$ space from the initial state to the final state. But remember that states in the Fermi sphere are occupied. If we want only slight change in $|\vec{k}|$ but large change in $\hat{k}$, the only thing we could do is assert:

Only electron states near the Fermi surface can be scattered.

If the electrons are not "swimming" in a field, they will be randomly sent to different parts of the Fermi sphere[22]. But with a non-zero drift velocity, thermodynamics starts to play. the scattered electrons will prefer the energy states with less energy, i.e. with smaller $|\vec{k}|$. That re-distribution will eventually cause the movement of Fermi sphere to stop. So, the total effect of field and scattering is a **shift** of Fermi sphere[23].

[21] Recall that the average energy of electrons are the same as ϵ_f in order of magnitude. Therefore, it is valid to compare with ϵ_f.

[22] Here we mean the surface, not the entire ball.

[23] Of course, at non-zero but finite temperatures the edge of Fermi sphere will be blurred due to the FD distribution.

Thermal conductivity and Wiedemann-Franz Law As we are using the same mathematical treatment, we may borrow the formular derived for phonons,

$$\kappa = \frac{1}{3}C\bar{v}\bar{l} = \frac{1}{3}\frac{\pi^2}{2}nk_\mathrm{B}\frac{T}{T_f}v_f\bar{l} = \frac{1}{3}\frac{\pi^2}{2}nk_\mathrm{B}\frac{2k_\mathrm{B}T}{m^*v_f^2}v_f^2\tau = \frac{\pi^2 nk_\mathrm{B}^2 T\tau}{2m^*}. \tag{7.60}$$

We used $\bar{v} = v_f$ here because only electrons near the surface of Fermi sphere can be scattered. Since $\tau \propto 1/T$, thermal conductivity is roughly constant with temperature.

At room temperature, electron thermal conductivity swamps phonon conductivity, therefore metals are generally better at conducting heat compared to insulators. At high temperatures, phonons dominate scattering.

Note that within the free electron model, the ratio $\kappa/\sigma T$ is a universal constant

$$\frac{\kappa}{\sigma} = \frac{\pi^2 k_\mathrm{B}^2 nT\tau/3m^*}{ne^2\tau/m^*} = \frac{\pi^2}{3}\frac{k_\mathrm{B}^2 T}{e^2} \quad\Rightarrow\quad L = \frac{\kappa}{\sigma T} = \text{const}, \tag{7.61}$$

where L is called the *Lorentz number*. This agrees well with experimental data, except at very low temperatures when different phonons related to the N and U scattering processes have excitation probabilities.

7.5.6 Where does the free electron model break down

Consider the existance of a magnetic field $\vec{B}$ perpendicular to the electric field, charge will build up on the sides of conductors to form an extra electric field $\vec{E}_H$ that balances out the magnetic field[24]. This is called the *Hall effect*.

Along the direction of $\vec{E}_H$,

$$m^*\frac{\mathrm{d}\vec{v}}{\mathrm{d}t} = -e\vec{E}_H - e\vec{v}\times\vec{B} = 0 \tag{7.62}$$

at equilibrium. Therefore

$$\vec{E}_H = \vec{B}\times\vec{v} = -\frac{1}{ne}\vec{B}\times\vec{j} = R_H\vec{B}\times\vec{j}, \tag{7.63}$$

where $R_H = -1/ne$ is the *Hall coefficient*. It seems to be a constant for any material, but actually it depends on a lot of things, temperature [25], material, etc.

Comparing the ratio of measured Hall coefficients [2] by looking at the ratio

$$\frac{n}{n_\mathrm{atom}} = \frac{1}{n_\mathrm{atom}eR_H} \tag{7.64}$$

where n_atom is the number of electrons per atom. Theory worked pretty well again for alkaline metals, but not for others, not even for alkaline earth metals such as Be. That suggests we need a more sophisticated theory that takes account of interactions with the periodic lattice.

Actually, the Hall effect is still not fully understood. There have been textbooks and articles discussing on different types of Hall effect, but experiments are still suggesting more. Maybe it is a good topic to look into, but we will need higher level of knowledge.

[24] Since $\vec{E}_H \parallel \vec{v}\times\vec{B} \propto \vec{E}\times\vec{B}$, it's easy to draw a diagram for this.

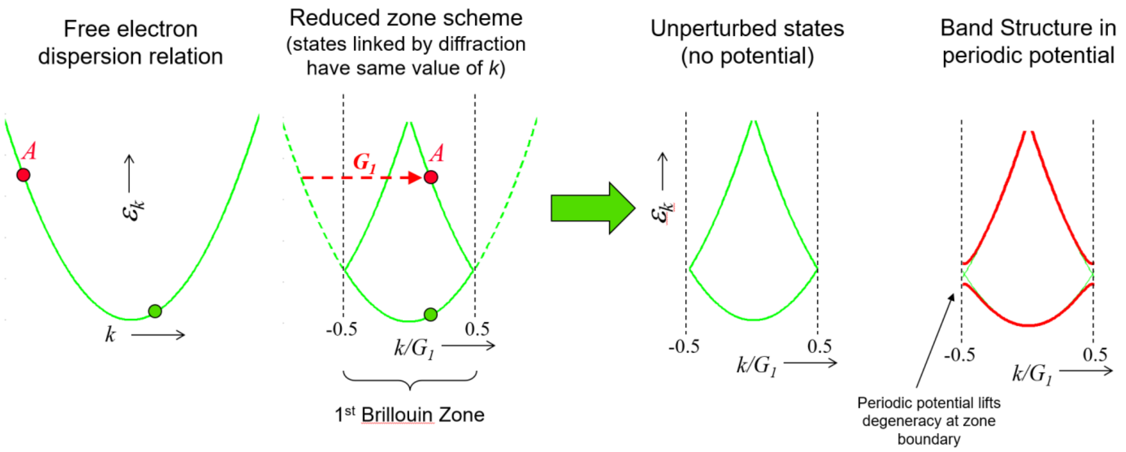

Figure 5. The electron dispersion relationship in the nearly free electron model. G_I is the lattice wavevector $2\pi/a$ where a is the distance between lattice points. Of course there are higher-energy backfoldings, but they are not shown in this diagram.

7.6 The Nearly Free Electron Model

So far, we have only considered insulators and metals with free electrons inside. It's true that we were successful at describing the properties of materials with the presence of electric field at low temperatures. But that is far from enough. We have already seen that once we apply a magnetic field the model breaks down. So we need to put a little bit more of the system, adding back in lattice periodicity.

For electrons, we denote their dispersion relationship using the function $\epsilon_k(k)$, where k is the wavenumber of the electron. For free electrons, the energy is simply kinetic energy

$$\epsilon_k = \frac{\hbar^2 k^2}{2m}, \tag{7.65}$$

which gives a quadratic function. If we add periodicity and introduce the Brillouin zones. The dispersion curve outside has to backfold into the first BZ, just as we did for phonons. Again, once potential is introduced, the connected dispersion curves will break up, as shown in Fig. 5.

Then, just write the full Hamiltonian out, and think about what we can do to simplify the model. Suppose we have only one kind of nuclei with charge Z and mass M at positions $\vec{R}_n$, valance electrons with mass m are at positions $\vec{r}_i$, the Hamiltonian would be [26]

$$\begin{aligned} H = &-\sum_i \frac{\hbar^2}{2m}\frac{\partial^2}{\partial \vec{r}_i^2} - \sum_n \frac{\hbar^2}{2M}\frac{\partial^2}{\partial \vec{R}_n^2} + \frac{e^2}{4\pi\epsilon_0^2}\sum_{i\neq j}\frac{1}{|\vec{r}_i - \vec{r}_j|} \\ &- \frac{Ze^2}{4\pi\epsilon_0}\sum_{i,n}\frac{1}{|\vec{r}_i - \vec{R}_n|} + \frac{Z^2e^2}{4\pi\epsilon_0}\sum_{n\neq m}\frac{1}{|\vec{R}_n - \vec{R}_m|}. \end{aligned} \tag{7.66}$$

We apply three approximations to the Hamiltonian:

- *Born-Oppenheimer (adiabatic) approximation*: We separate the motion of electrons and nuclei, which have very different masses. When studying the motion of electrons, we assume that the nuclei are stationary, and when studying the motion of nuclei, we do not need to consider the distribution of electrons. Since we are studying electrons, this allows us to drop the second term, which is the kinetic energy of the nuclei, and treat the last term (inter-nuclei potential) as constant.

- *Single-electron approximation*: The interaction felt by an electron by other electrons is approximated as a constant, time-independent potential field. This allows us to write the inter-electron interaction term as $\sum_i v_e(\vec{r}_i)$, the sum of potential felt by each electron.

- *Periodic potential*: In a crystal, the total potential field felt by a single electron has the same periodicity as the lattice.

and it becomes

$$H = \sum_i \left\{ -\frac{\hbar^2}{2m}\frac{\partial^2}{\partial \vec{r}_i^2} + v_e(\vec{r}_i) - \frac{Ze^2}{4\pi\epsilon_0}\sum_n \frac{1}{|\vec{r}_i - \vec{R}_n|} \right\} = \sum_i \left\{ -\frac{\hbar^2}{2m}\frac{\partial^2}{\partial \vec{r}_i^2} + V(\vec{r}_i) \right\}, \quad (7.67)$$

where we have an expression for the single-electron Hamiltonian in the brackets. Now we cannot go further without any hints, and here's where *Bloch's theorem* comes in.

7.6.1 Bloch's theorem

Bloch's theorem is purely from the translational symmetry, which suggests that electron probability density must have the same periodicity as the lattice. It has two equivalent expressions:

- The solutions of Schrödinger equation in a periodic lattice must have the form $\psi_k(\vec{r}) = u_k(\vec{r})\mathrm{e}^{\mathrm{i}\vec{k}\cdot\vec{r}}$, where $u(\vec{r})$ is a periodic function that suggests the periodicity of the lattice.

- In a periodic potential $V(\vec{r}) = V(\vec{r}+\vec{R})$, wavefunction of eigenstates must obey $\psi(\vec{r}+\vec{R}) = \mathrm{e}^{\mathrm{i}\vec{k}\cdot\vec{R}}\psi(\vec{r})$.

That is to say, if $\vec{R}$ is a lattice vector in real space, moving by $\vec{R}$ will change the wavefunction only by a phase factor $\mathrm{e}^{\mathrm{i}\vec{k}\cdot\vec{R}}$.

Functions suggested by Block's theorem are called *Bloch's functions*, and electrons described using this framework are called *Bloch's electrons*.

Expand u_k into Fourier series, for a one-dimensional lattice

$$u_k(x) = \sum_n C_{k,n}\frac{1}{\sqrt{A}}\mathrm{e}^{\mathrm{i}mGx}, \qquad G = \frac{2\pi}{a}, \quad (7.68)$$

where A is a dimensionless factor for normalisation. The overall wavefunction then takes the form

$$\psi_k(x) = \sum_{n=-\infty}^{\infty} C_{k,n}|\phi_{k,n}\rangle, \qquad |\phi_{k,n}\rangle = \frac{1}{\sqrt{A}}\mathrm{e}^{\mathrm{i}(k+nG)x}, \quad (7.69)$$

the basis states are shown in Fig. 6.

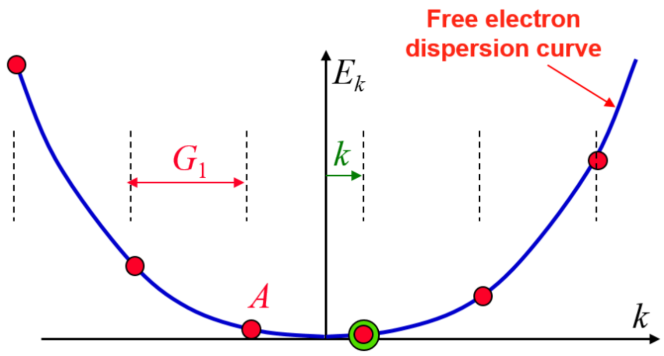

Figure 6. In the dispersion relation diagram, the $|\phi_{k,n}\rangle$ basis states are separated by G along the k-axis, and the energies lie on the free-electron curve because these are plane wave solutions.

To get the energy eigenvalues, we begin by construct the matrix elements of H,

$$\begin{aligned}
H\psi(x) &= \epsilon\psi(x) \\
H\sum_m C_{k,m}|\phi_{k,m}\rangle &= \epsilon\sum_p C_{k,p}|\phi_{k,p}\rangle \\
\sum_m C_{k,m}H|\phi_{k,m}\rangle &= \epsilon\sum_p C_{k,p}|\phi_{k,p}\rangle \\
\sum_m C_{k,m}\langle\phi_{k,n}|H|\phi_{k,m}\rangle &= \epsilon\sum_p C_{k,p}\langle\phi_{k,n}|\phi_{k,p}\rangle
\end{aligned}$$

Now we have H_{mn} in the equation, which suggests

$$\sum_m H_{mn}C_{k,m} = \epsilon C_{k,n} \quad \Rightarrow \quad \mathbf{H}\vec{C} = \epsilon\vec{C}, \tag{7.70}$$

where $\vec{C}$ is a vector of the coefficients of the basis-states. Therefore the eigenvectors of H gives the **set** of coefficients $C_{k,n}$ of the basis functions in the solutions of the Schrödinger equation.

The matrix elements H_{mn} can be calculated directly by the inner product, and expand $V(x)$ using Fourier series with coefficients V_p.

$$\begin{aligned}
H_{nm} &= \langle\phi_{k,n}|H|\phi_{k,m}\rangle \\
&= \int_0^A \frac{1}{\sqrt{A}}\mathrm{e}^{-\mathrm{i}(k+nG)x}\left(-\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2} + V(x)\right)\frac{1}{\sqrt{A}}\mathrm{e}^{\mathrm{i}(k+mG)x}\,\mathrm{d}x \\
&= \frac{1}{A}\int_0^A \frac{\hbar^2(k+mG)^2}{2m}\mathrm{e}^{-\mathrm{i}(n-m)Gx}\,\mathrm{d}x \\
&\quad + \frac{1}{A}\int_0^A \mathrm{e}^{-\mathrm{i}(n-m)Gx}\left[\sum_p \frac{V_p}{2}\left(\mathrm{e}^{-\mathrm{i}pGx} + \mathrm{e}^{\mathrm{i}pGx}\right)\right]\,\mathrm{d}x \\
&= \frac{\hbar^2(k+mG)^2}{2m}\delta_{mn} + \sum_p \frac{V_p}{2}\left(\delta_{n,m+p} + \delta_{n,m-p}\right),
\end{aligned} \tag{7.71}$$

$$\psi_+(x) = \frac{1}{2}\left[\left|\phi_{G_1/2}\right\rangle + \left|\phi_{-G_1/2}\right\rangle\right] \propto \cos\left(\frac{\pi}{a}x\right)$$

$$\psi_-(x) = \frac{1}{2}\left[\left|\phi_{G_1/2}\right\rangle - \left|\phi_{-G_1/2}\right\rangle\right] \propto \sin\left(\frac{\pi}{a}x\right)$$

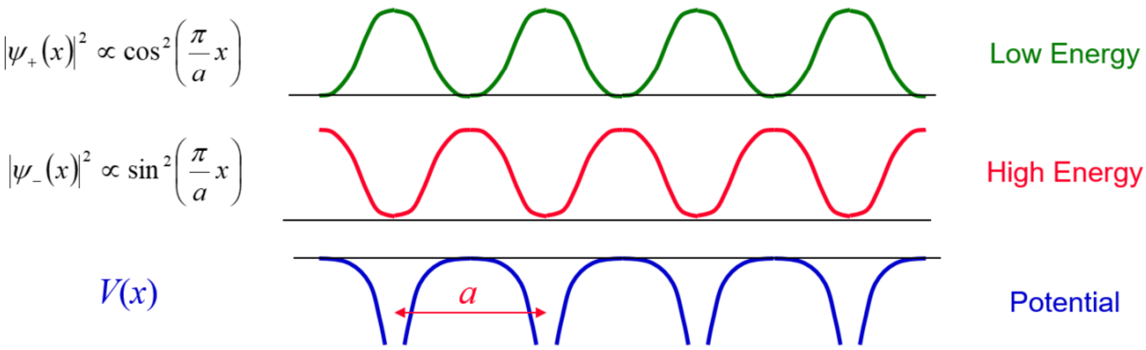

Figure 7. Suppose we only have two $|\phi_{n,k}\rangle$ states separated by G. Electrons will prefer the lower-energy state. When the two states contribute equally, they superpose to form standing waves. The superposition can be constructed in two ways as shown. The two new states have different energies, so the degeneracy in the dispersion relationship is lifted; the two branches split apart in energy.

where the diagonal matrix elements are given by kinetic terms, and off-diagonal elements are given by potential energy expansion. Note that this is a matrix of rank ∞, therefore in reality the only way to investigate the matrix is time-independent perturbation theory, where we treat $V(x)$ as perturbation.

If we take the two-states approximation shown in Fig. 7, where

$$\psi_k(x) = C_{k,0}|\phi_{k,0}\rangle + C_{k,-1}|\phi_{k,-1}\rangle,$$

the matrix would be 2×2, which is a lot easier to solve,

$$\mathbf{H}\vec{C} = \begin{pmatrix} \frac{\hbar^2(k-G)^2}{2m} & V_1/2 \\ V_1/2 & \frac{\hbar^2 k^2}{2m} \end{pmatrix} \begin{pmatrix} C_{k,-1} \\ C_{k,0} \end{pmatrix} = \epsilon_k \begin{pmatrix} C_{k,-1} \\ C_{k,0} \end{pmatrix}.$$

Rearrange and make determinant zero, we can find the eigenvalues of the system

$$\epsilon_k = \frac{1}{2}(E_{k,-1} + E_{k,0}) \pm \sqrt{\frac{1}{4}(E_{k,-1} - E_{k,0})^2 + \left(\frac{V_1}{2}\right)^2}. \tag{7.72}$$

Note that we only have V_1 in the matrix and energy eigenvalues, that's because the range of n and m are limited. Higher p Fourier components link states at higher energies, which gives energy gaps of V_p at $k = pG/2$. Similar to phonons, we know see that electrons have a similar **band structure** in principle, as shown in Fig. 8.

7.6.2 Electron band structure

The band structure intuitively suggests that the Fermi sphere has gaps inside, as $\epsilon \propto |k|^2$. We may take a closer look at how much the band is filled by looking at how many valance

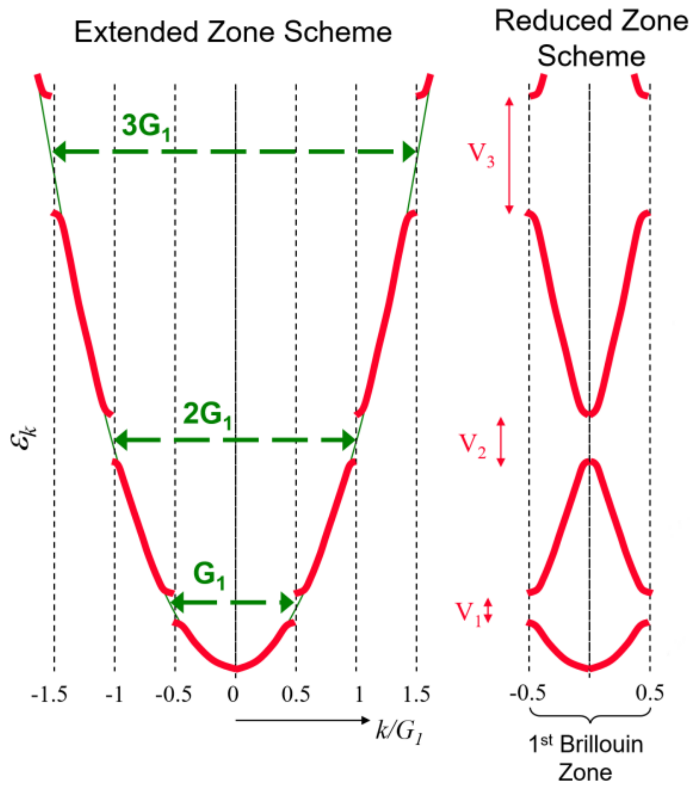

Figure 8. The origin of electronic band structure. Of course, electrons fill the band up to the Fermi energy ϵ_f, and the level of filling of the bands determines whether a material is a conductor (partially filled), insulator (full, because there are no near-surface states available for the Fermi sphere to shift) or a semiconductor (full, but low band gap).

electrons we have for each lattice point. For instance, if we have one electrons per atom, we will have the lowest band half-full because each state can hold up to two electrons.

But this analysis would suggest that all divalent elements (two valance electrons per atom) are insulators, which is clearly not the case. Why? Because we forgot that we are in 1D. Things in higher dimensions are more complicated.

We start by adding only one dimension, because it is easy to think of, and we do have a lot of materials whose k-space only has two dimensions, see [27] and related works.

In a 2D metal, the free electron model will give a parabola,

$$\epsilon = \frac{\hbar^2}{2m}\left(k_x^2 + k_y^2\right), \tag{7.73}$$

and there are of course no bands. States are filled up to the Fermi energy ϵ_f, which forms a circular contour projected to the k_x-k_y plane. However, if we add a weak potential as perturbation, small energy gap is formed at the zone boundary. The shape of the boundary is determined using the standard rule for Wigner–Seitz cell, but we will just use a square as an example.

Now, part of the circle would go outside the square, forming band gaps and would potentially fill in higher bands, leading to *energy overlap* between bands. But if the periodic

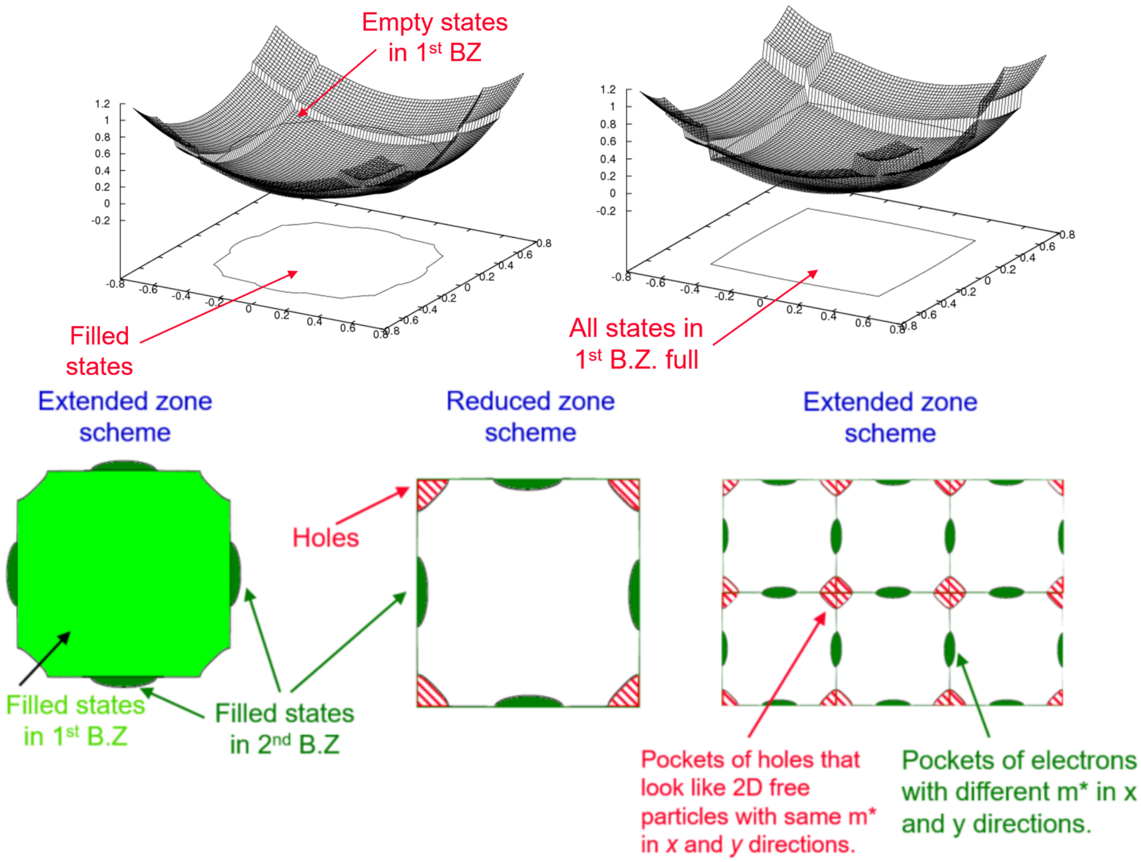

Figure 9. Band structure of a divalent conducting material under weak potential (top left) and high potential (top right). The filled states are projected to the k subplane. At the band gap, energies vary with position around the BZ edge, so the energy-wavevector relation depends on the direction of the wavevector you choose. The region of holes are indicated in the bottom figure.

field is large for some material, it will remain an insulator because the band gap is larger, and there is no overlap in energies between bands.

Since we are able to identify conductors, we may move on and consider how does the Fermi sphere move in those conductors using our *nearly* free electrons model.

7.6.3 Bloch oscillations and effective mass

For a partially filled band under relatively low electric field, the behaviour is similar to that of free electron model. But if the **electric field is strong** and the **scattering processes are weak**, the occupied states can move continuously through k-space. With backfolding[25], it looks like an oscillation inside the first BZ. This is called *Bloch oscillations*, and can be seen both in condensed matter [28–30] and in optical lattices [31–33].

Using the group velocity idea, we can work out the electron effective mass quantitatively.

$$f = m^* \frac{\mathrm{d}v}{\mathrm{d}t} = \hbar \frac{\mathrm{d}k}{\mathrm{d}t}, \quad \text{and} \quad v = \frac{\mathrm{d}\omega}{\mathrm{d}k} = \frac{1}{\hbar}\frac{\mathrm{d}\epsilon}{\mathrm{d}k}, \tag{7.74}$$

[25] Inversion of direction of the group velocity.

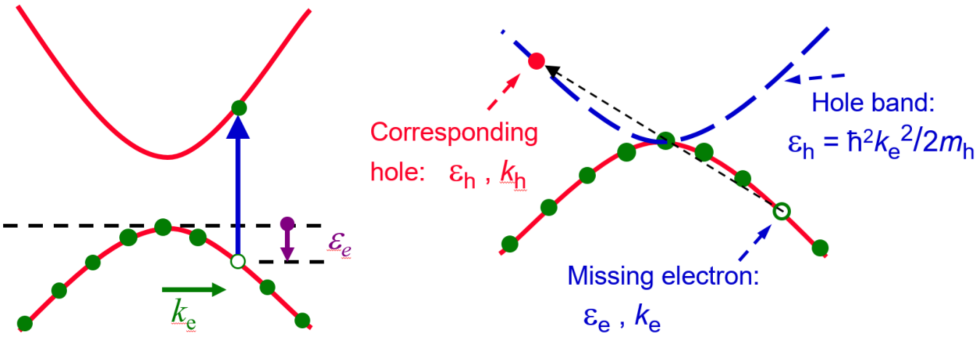

Figure 10. We define holes such they carry energy and momentum representing the almost full valence band.

which gives,

$$\frac{\mathrm{d}v}{\mathrm{d}t} = \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\mathrm{d}\omega}{\mathrm{d}k}\right) = \frac{1}{\hbar}\frac{\mathrm{d}^2\epsilon}{\mathrm{d}k^2}\frac{\mathrm{d}k}{\mathrm{d}t} \quad \Rightarrow \quad m^* = \hbar^2\left(\frac{\mathrm{d}^2\epsilon}{\mathrm{d}k^2}\right)^{-1}. \tag{7.75}$$

The second derivative inside means that the effective mass depends on the band structure at the particular band and particular wavevector k. Depending on the position of ϵ_f, **more than one effective mass** can contribute to conduction in dimensions $D > 1$.

7.6.4 Holes and Hall effect revisited

Holes are effectively charge-carriers that were invented for those whose bands are almost full, such as two electrons per atom with weak potential. Note that holes move in the opposite direction of electrons.

Of course, for holes to be present, we need to transfer at least some electrons into the band with higher energy. The energy required to do that is

$$\Delta\epsilon = E_{\text{gap}} + \frac{\hbar^2 k^2}{2m_e} + \frac{\hbar^2 k^2}{2m_h}, \tag{7.76}$$

where m_h is the mass of the hole. It can be defined, using a combination of energy and momentum, as shown in Fig. 10.

Note that for ϵ_h to be negative we require a negative m^*, and it is satisfied because at the top of a band, the band curvature is certainly negative. That makes a hole quasiparticle instead of antiparticle. But for simplicity, we still consider a hole with positive charge, mass and energy.

Now we move on to see how it can affect our pridictions on Hall effect. Since both electrons and holes contribute to conduction, the current density is

$$\vec{j} = (\sigma_e + \sigma_h)\vec{E}, \quad \text{where} \quad \sigma_e = \frac{n_e e^2 \tau_t}{m_e^*}, \quad \sigma_h = \frac{n_h e^2 \tau_h}{m_h^*}. \tag{7.77}$$

It can be shown that[26] the hall coefficient can be written in terms of the number density

[26] I haven't got time to prove that.

and mobility of both electrons and holes,

$$R_H = \frac{-n_e\mu_e^2 + n_h\mu_h^2}{e(n_e\mu_e + n_h\mu_h)^2}. \tag{7.78}$$

That explains the possibility of negative Hall coefficient, because when $n_e \sim n_h$, the species with higher mobility will dominate the Hall effect.

As for the comparison of effective mass, it can be done using *cyclotron resonance*. In a B field, charge particles perform circular orbits around the field lines, with frequency $\omega = eB/m$, this is typically in the radio waves region. Therefore we can shoot the sample using radio waves, and at certain frequencies strong absorption will occur, indicating resonance. From the frequencies, the effective masses can be determined, as e is a constant and B can be measured directly [34, 35]. Of course, it needs high ω, low temperature and strong B fields.

Now we could describe both conductors and insulators quite well, and we shall move on to semiconductors and semiconductor devices.

7.7 Semiconductors and Semiconductor Devices

Before everything, we must ask: What defines a semiconductor, because we haven't discussed anything about that. In simple words, we have a **full band at** $T = 0$, but the band gap between the valence band and the conductance band is small enough to be **crossed by thermal excitation** at $T > 0$. So, the conductance is intrinsically temperature-dependent.

We can find materials in nature that has those properties, which are *intrinsic* semiconductors, such as Si or Ge. But there are *extrinsic* semiconductors made from *doping* the material with suitable impurities.

7.7.1 Doping

Doping refers to replacing a few of the semiconductor atoms (valence 4) with impurity atoms of a different valence (usually 3 or 5). Note that this will also change the mechanical and thermal properties of the material, which was discussed in the IA materials course.

Consider adding P doping to Si. We can regard P as and ion P^+ plus an electron e^-. The ion has core shells full and same number of valence electrons as Si, therefore the valence band is full and the conduction band is still empty. Everything looks the same before adding the electron, except that the energy of states changed slightly.

The extra electron must sit in the conduction band, but don't need to sit next to the ion. However, there is an attractive interaction between the electron and the ion due to Coulomb force. Hence, the combination of the P atom and the electron can exist in two states:

- An ionised state, where the electron and positive ion are separated.
- A **bound (hydrogenic) state**, where the electron and positive ion form a hydrogen like structure.

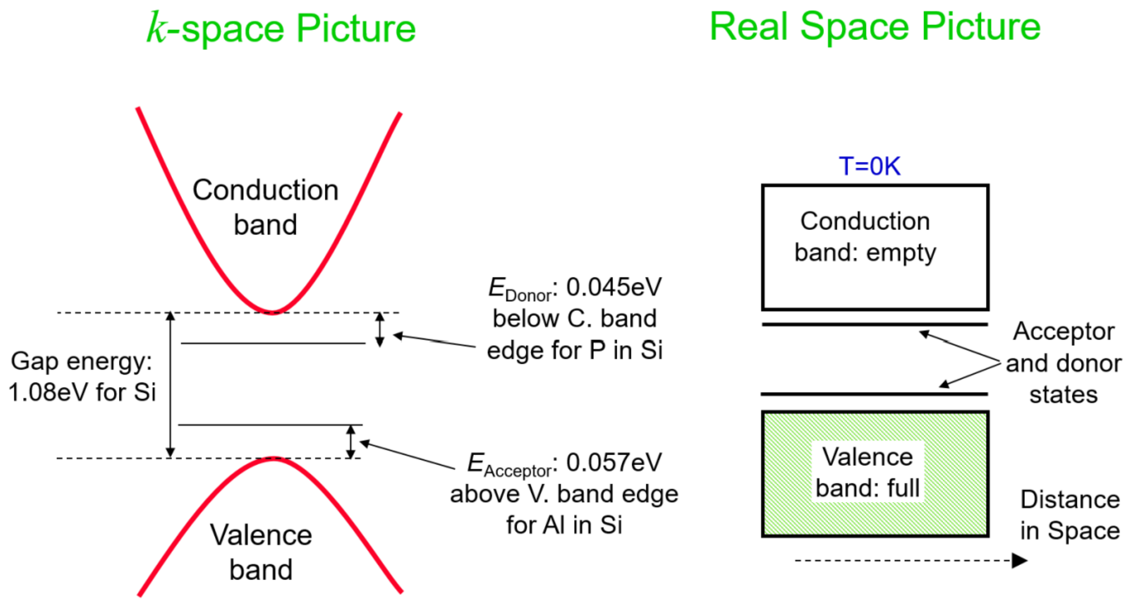

Figure 11. In Si, the donor states for n-type doping and the acceptor states for p-type doping stays between the valence and conduction bands. For an n-type semiconductor, at low temperatures, donor states are full and conduction band is empty. Then electrons in the donor states are excited thermally into the conduction band, turning the material into a conductor. For a p-type semiconductor, the valence band is full and acceptor states empty at low temperatures. As temperature increase, acceptor states are ionised (occupied by electrons) and holes goes down into the valence band, making the material a conductor.

The ionised states contributes to the conductivity of the material because electrons can move freely in the material. But it's harder to analyse what would happen for the bound state.

If the electron sits in the hydrogenic states, electrostatic potential results in binding energy between the electron and the ion. Therefore, the electron would be in a k-independent state with minus energy[27] given by

$$E_n = -\frac{m^* e^4}{32\pi^2\epsilon^2\epsilon_r^2\hbar^2 n^2}, \qquad r_n = \frac{4\pi\epsilon_0\epsilon_r\hbar^2 n^2}{m^* e^2}. \tag{7.79}$$

For each defect state created, one state is lost from the band it was created from.

Since the energy is too low, at room temperature it is very likely that the bound state electrons are almost all excited to the ionised state. Normally, the bound states won't contribute to conductivity, However, note that the radius of the bound state may be larger than the atomic spacing in the lattice. So, if we have a large number of bound states overlapping in space, and generating an *impurity band* which will contribute to the *superconductivity* of the material.

Up to now we only discussed substitutes with more electrons, which corresponds to the **n**-type doping, but haven't discussed on the **p**-type doping which will form acceptor

[27] A straight line below $\epsilon = 0$ in the first BZ.

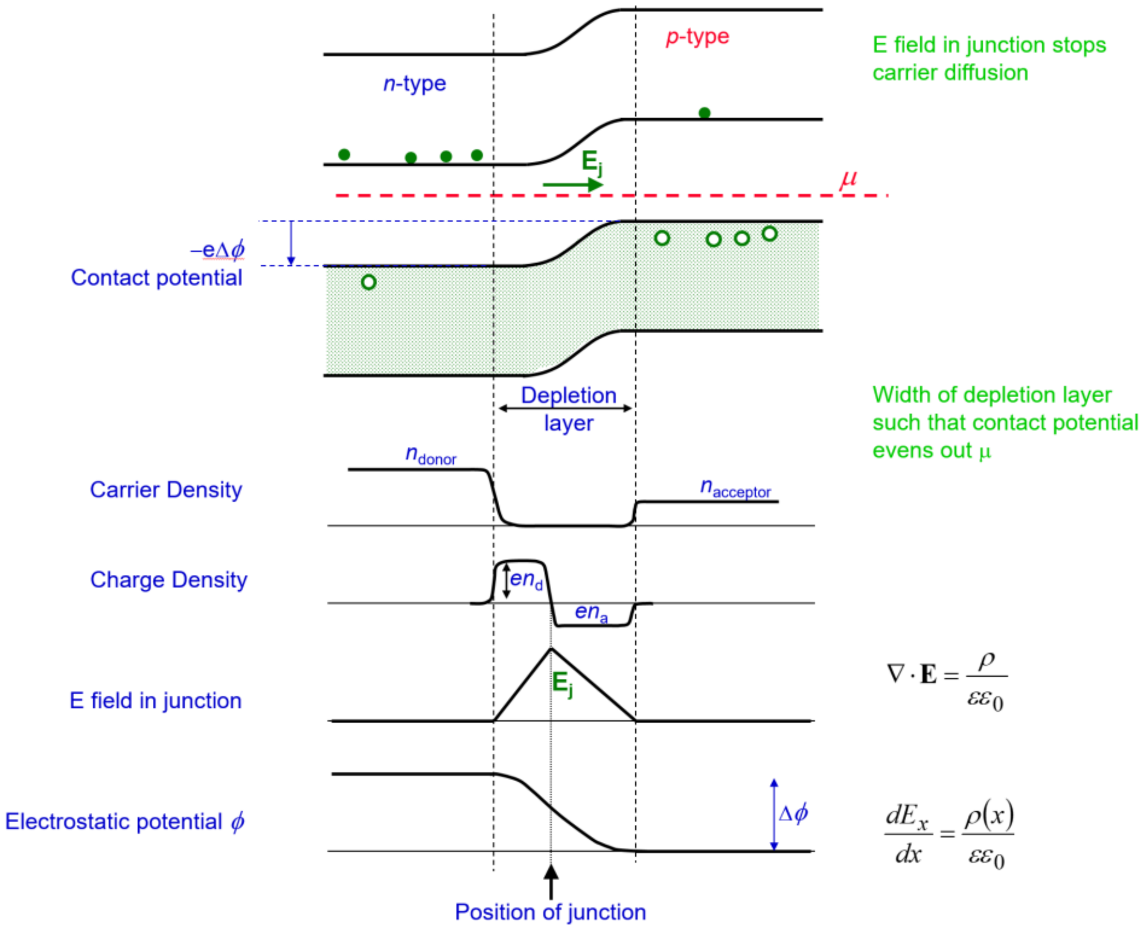

Figure 12. The picture used to analyse what happens in a p-n junction. The upper part shows the bands, shifted to match chemical potential, and the electric field in the depletion layer stops carrier diffusion. Down below there are functions such as carrier and charge densities, field and potential w.r.t position. The junction lies inside the depletion layer on the charge neutral (highest field) point.

states. A good example is the Al doping in Si. It will result in a hole that can be either ionised or bound, as before.

In general, we have our valence band and conduction band, where the acceptor and donor states stay between the bands. And we have a band gap relatively low (< 2 eV) between bands. As shown in Fig. 11, the thermal excitation is due to a shift in the chemical potential of the doped semiconductor. As temperature increases, **chemical potential moves towards the middle of the gap**[28]. But at high T the number of thermally excited carriers increases, so donor states becomes less important.

Now we can build something called p-n junction which is central to electronic devices.

[28] Recall that μ is the energy level where we have 50% occupation.

7.7.2 P-N junction

The basic idea is to enable carries to move between the two types of material, and make use of the *depletion region* formed at the interface. So, electrons diffuse from n to p, and holes diffuse from p to n. The diffusing carriers leave their associated ions behind, and the charge separation produces an electric field at the junction that stops further carrier diffusion. This gives rise to an electrostatic potential that balances the difference in μ[29]. This is summarised in Fig. 12, a rather complicated picture.

We can now think about what happens to our junction as we apply a bias (field) to it. We can either apply a *forward bias* (field from p to n) or a *reverse bias* (from n to p). For a forward bias the potential difference would force the majority carriers to move towards the junction (*recombination current*), leading to good conduction. Instead, a reverse bias would increase width of the depletion layer (*generation current*), and reduce conductivity.

Actually, in the n-type semiconductor μ increases to $\mu + eV$ is we add a bias V forward. the Fermi distribution becomes

$$p_V(\epsilon) = p_0(\epsilon) \exp\left[\frac{eV}{k_\mathrm{B}T}\right]. \tag{7.80}$$

Note that we have assumed that $\epsilon \gg \mu$ to do the estimate

$$p_0(\epsilon) \simeq \exp\left[-\frac{\epsilon-\mu}{k_\mathrm{B}T}\right].$$

The current-voltage relation of a p-n diode can then be obtained using the expression for the increase in carrier population with bias. For no bias, the recombination current balances the generation current, therefore the curve passes through $(0,0)$. The bias does not change the generation current, but does change the recombination current because the number of electrons in the n-type region that can cross the junction varies with $\exp[eV/k_\mathrm{B}T]$. Therefore, current varies as

$$I = I_0\left[\exp\left(\frac{eV}{k_\mathrm{B}T} - 1\right)\right], \tag{7.81}$$

where I_0 characterises the generation current, and acts as the reverse current due to thermally excited minority carriers when $V < 0$. For holes, exact same argument applies.

This is indeed much different from the diodes in reality, and there is much more to talk about, such as for Zener diodes [36–38]. We shall discuss them in the following subsection.

7.7.3 Semiconductor devices

Even for the simplest diodes, we can make it useful by combining them in suitable order, such as the bridge rectifier in Fig. 13. This is because diodes has good conduction with forward bias and poor conduction with reverse bias, as discussed previously.

[29]Note that we need to match the chemical potential, as required by charge neutrality. Often we do that by rising the energy of p-type region entirely until p and n share the same μ.

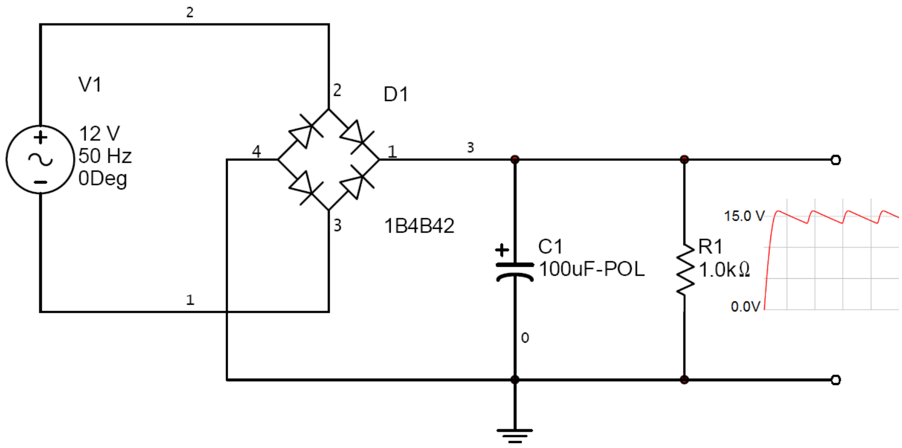

Figure 13. A typical bridge rectifier that preserves all current (Wikipedia). A fast way to memorise it is: All diodes points to the right.

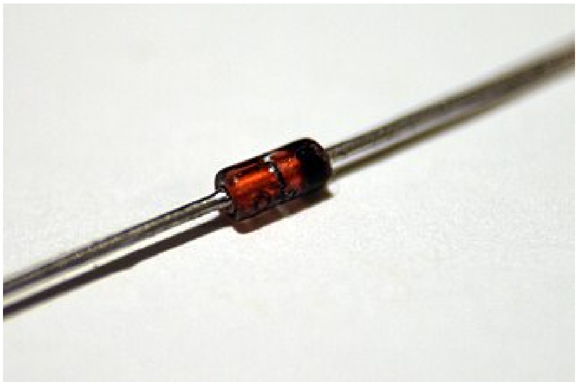

Figure 14. The picture of a Zener diode to be used on PCB boards (Wikipedia). Zener diodes often look pretty with the copper-like color and transparent coating, while avalanche diodes usually have ugly black coating and white lines on it (indicating its properties).

Zener diode We can form a *Zener diode*[30], shown in Fig. 14, by carefully choosing our material. With sufficient **reverse** bias, a p-n junction can break down and conduct in the reverse direction, which we do not want it to happen. What we can do is to let *Zener breakdown* happen first by heavily doping[31] our materials.

Zener breakdown is caused by tunnelling at large reverse bias (typically above 3 V). The large bias causes the energy at the top of the valence band in the p-type material to overlap the bottom of the conduction band in the n-type. As one can imagine, that causes electrons to tunnel towards the n-type, and fill in its conduction band with correct energy.

[30] Often called *regulator* diode, because it stablises the voltage across it. So it can regulate the circuit by acting as a voltage reference.

[31] Heavy doping puts the chemical potential near the band edge and reduces the voltage needed for breakdown, so Zener breakdown occurs before other breakdowns.

Avalanche diode Another kind of diode is called *avalanche diode*, which also breaks down under **reverse** bias. But it is caused by successive generation of carriers. Under reverse bias, thermally excited carrier within the junction gain energy between collisions, and create further electron-hole pairs, which gain energy and so form even more carriers. As a result, large reverse current can flow in the junction, and a lot of heat can be dissipated, leading to a damage of the diode.

The avalanche diode can also be used as a voltage reference, but it is more commonly used for protecting the circuit against high voltages. It can even be used as a single photon detector, as an incoming photon could potentially trigger the avalanche in a device held very near to that point[32]. This is similar in principle to the photomultiplier tube, where the avalanche is made possible by applying high voltage between each dynode pairs.

Light Emitting Diodes (LED) A LED is a p-n junction that is optimised for the production of photons during carrier recombination. It is hard to say who developed the first light-emitting diode, but its history can trace back to the early 1900s.

To emit light, we need the electron to drop from the conduction band of the n-type to combine with the holes from the conduction band of p-type, which needs a forward bias. The recombination increases entropy, therefore releases energy, in the form of photons[33]. The size of the band gap determines the wavelength of the photons emitted.

Since E and p (or k) are both conserved, light emission needs a relatively little change in k for the electrons, corresponding to direct gap transition. If there is an indirect gap transition, which changes k dramatically, the extra momentum must be supplied by something else, such as a phonon.

One could easily imagine that it would be hard to emit a photon of relatively high energy (short wavelength, 'blue'), because it needs a wide band gap. But unluckily this problem was solved [39] by Japanese physicists and honoured 2014 Nobel Prize in Physics.

Light Amplification by Stimulated Emission of Radiation (LASER) LED can emit light of a certain wavelength, but the light is not generally coherent. To obtain coherent light, the electrons needs to be driven[34] up and down in energy level by light. That give rise to a *population inversion* of energy states. Of course, this needs very heavy doping and a strong forward bias, so carriers are injected into the other side of the junction to maintain the population inversion. Also, to maintain the light, we need a cavity similar to Fabry-Perot etalon and a find control system to stablise the wavelength of the light. The detailed mechanism will be explained in Part II AQP.

Solar cells Other than emitting energy, p-n junctions can absorb energy as well. If a photon comes into the junction, carrier pairs are created by the energy, and swept out of the junction by the intrinsic electric field inside, creating a photo-current, I_L. However,

[32] A lot of doping, of course.
[33] Other forms of energy release can occur, but photon is the most favorable process.
[34] "pumped"

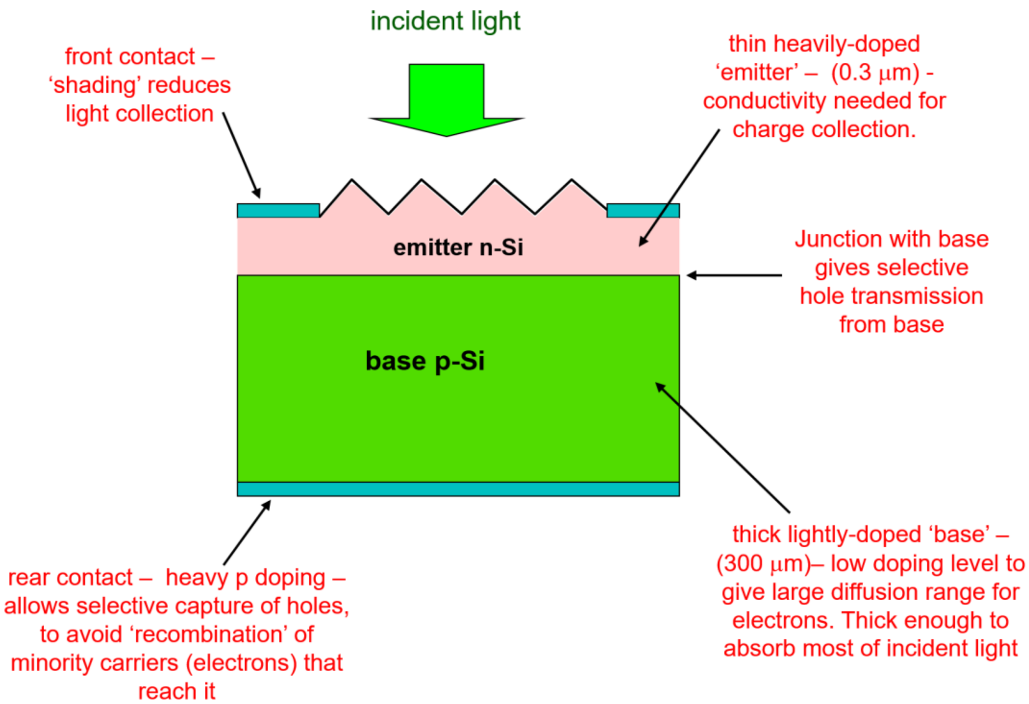

Figure 15. A sketch of the modern silicon solar cell. Efficiencies of larger than twenty percent can be achieved.

accumulation of charges at each side of the junction will act as forward bias, and some of the current leaks back again.

Equivalently, it can be modelled using a current source in parallel with diode and a shunt resistor (leakage). The output current will therefore be

$$I = I_L - I_{\text{diode}} - I_{\text{leakage}}, \tag{7.82}$$

which is a equation of no help at all.

Of course, there are other countless semiconductor devices. Mobile phones and computers are products that perform logic operations based on triodes and other stuff. But these are non-examinable.

The End.

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